UM10360

Chapter 14: LPC17xx UART0/2/3

Rev. 01 — 4 January 2010 User manual

1. Basic configuration

The UART0/2/3 peripherals are configured using the following registers:

1. Power: In the PCONP register (Table 4–46), set bits PCUART0/2/3.

   Remark: On reset, UART0 is enabled (PCUART0 = 1), and UART2/3 are disabled (PCUART2/3 = 0).
2. Peripheral clock: In the PCLKSEL0 register (Table 4–40), select PCLK_UART0; in the PCLKSEL1 register (Table 4–41), select PCLK_UART2/3.
3. Baud rate: In register U0/2/3LCR (Table 14–278), set bit DLAB =1. This enables access to registers DLL (Table 14–272) and DLM (Table 14–273) for setting the baud rate. Also, if needed, set the fractional baud rate in the fractional divider register (Table 14–284).
4. UART FIFO: Use bit FIFO enable (bit 0) in register U0/2/3FCR (Table 14–277) to enable FIFO.
5. Pins: Select UART pins through the PINSEL registers and pin modes through the PINMODE registers (Section 8–5).

   Remark: UART receive pins should not have pull-down resistors enabled.
6. Interrupts: To enable UART interrupts set bit DLAB =0 in register U0/2/3LCR (Table 14–278). This enables access to U0/2/3IER (Table 14–274). Interrupts are enabled in the NVIC using the appropriate Interrupt Set Enable register.
7. DMA: UART0/2/3 transmit and receive functions can operate with the GPDMA controller (see Table 31–544).

2. Features

- Data sizes of 5, 6, 7, and 8 bits.
- Parity generation and checking: odd, even mark, space or none.
- One or two stop bits.
- 16 byte Receive and Transmit FIFOs.
- Built-in baud rate generator, including a fractional rate divider for great versatility.
- Supports DMA for both transmit and receive.
- Auto-baud capability
- Break generation and detection.
- Multiprocessor addressing mode.
- IrDA mode to support infrared communication.
- Support for software flow control.

3. Pin description

Table 268: UARTn Pin description

Pin	Type	Description
RXD0, RXD2, RXD3	Input	**Serial Input.** Serial receive data.
TXD0, TXD2, TXD3	Output	**Serial Output.** Serial transmit data.

4. Register description

Each UART contains registers as shown in Table 14–269. The Divisor Latch Access Bit (DLAB) is contained in UnLCR7 and enables access to the Divisor Latches.

Table 269. UART0/2/3 Register Map

Generic Name	Description	Access	Reset value[1]	UARTn Register Name & Address
RBR (DLAB =0)	Receiver Buffer Register. Contains the next received character to be read.	RO	NA	U0RBR - 0x4000 C000 U2RBR - 0x4009 8000 U3RBR - 0x4009 C000
THR (DLAB =0)	Transmit Holding Register. The next character to be transmitted is written here.	WO	NA	U0THR - 0x4000 C000 U2THR - 0x4009 8000 U3THR - 0x4009 C000
DLL (DLAB =1)	Divisor Latch LSB. Least significant byte of the baud rate divisor value. The full divisor is used to generate a baud rate from the fractional rate divider.	R/W	0x01	U0DLL - 0x4000 C000 U2DLL - 0x4009 8000 U3DLL - 0x4009 C000
DLM (DLAB =1)	Divisor Latch MSB. Most significant byte of the baud rate divisor value. The full divisor is used to generate a baud rate from the fractional rate divider.	R/W	0x00	U0DLM - 0x4000 C004 U2DLM - 0x4009 8004 U3DLM - 0x4009 C004
IER (DLAB =0)	Interrupt Enable Register. Contains individual interrupt enable bits for the 7 potential UART interrupts.	R/W	0x00	U0IER - 0x4000 C004 U2IER - 0x4009 8004 U3IER - 0x4009 C004
IIR	Interrupt ID Register. Identifies which interrupt(s) are pending.	RO	0x01	U0IIR - 0x4000 C008 U2IIR - 0x4009 8008 U3IIR - 0x4009 C008
FCR	FIFO Control Register. Controls UART FIFO usage and modes.	WO	0x00	U0FCR - 0x4000 C008 U2FCR - 0x4009 8008 U3FCR - 0x4009 C008
LCR	Line Control Register. Contains controls for frame formatting and break generation.	R/W	0x00	U0LCR - 0x4000 C00C U2LCR - 0x4009 800C U3LCR - 0x4009 C00C
LSR	Line Status Register. Contains flags for transmit and receive status, including line errors.	RO	0x60	U0LSR - 0x4000 C014 U2LSR - 0x4009 8014 U3LSR - 0x4009 C014
SCR	Scratch Pad Register. 8-bit temporary storage for software.	R/W	0x00	U0SCR - 0x4000 C01C U2SCR - 0x4009 801C U3SCR - 0x4009 C01C
ACR	Auto-baud Control Register. Contains controls for the auto-baud feature.	R/W	0x00	U0ACR - 0x4000 C020 U2ACR - 0x4009 8020 U3ACR - 0x4009 C020
ICR	IrDA Control Register. Enables and configures the IrDA mode.	R/W	0x00	U0ICR - 0x4000 C024 U12CR - 0x4009 8024 U3ICR - 0x4009 C024
FDR	Fractional Divider Register. Generates a clock input for the baud rate divider.	R/W	0x10	U0FDR - 0x4000 C028 U2FDR - 0x4009 8028 U3FDR - 0x4009 C028
TER	Transmit Enable Register. Turns off UART transmitter for use with software flow control.	R/W	0x80	U0TER - 0x4000 C030 U2TER - 0x4009 8030 U3TER - 0x4009 C030
FIFOLVL	FIFO Level register. Provides the current fill levels of the transmit and receive FIFOs.	RO	0x00	U0FIFOLVL - 0x4000 C058 U2FIFOLVL - 0x4009 8058 U3FIFOLVL - 0x4009 C058

[1] Reset Value reflects the data stored in used bits only. It does not include reserved bits content.

14.4.1 UARTn Receiver Buffer Register (U0RBR - 0x4000 C000, U2RBR - 0x4009 8000, U3RBR - 0x4009 C000 when DLAB = 0)

The UnRBR is the top byte of the UARTn Rx FIFO. The top byte of the Rx FIFO contains the oldest character received and can be read via the bus interface. The LSB (bit 0) represents the “oldest” received data bit. If the character received is less than 8 bits, the unused MSBs are padded with zeroes.

The Divisor Latch Access Bit (DLAB) in LCR must be zero in order to access the UnRBR. The UnRBR is always read-only.

Since PE, FE and BI bits correspond to the byte sitting on the top of the RBR FIFO (i.e. the one that will be read in the next read from the RBR), the right approach for fetching the valid pair of received byte and its status bits is first to read the content of the U0LSR register, and then to read a byte from the UnRBR.

Table 270: UARTn Receiver Buffer Register (U0RBR - address 0x4000 C000, U2RBR - 0x4009 8000, U3RBR - 04009 C000 when DLAB = 0) bit description

Bit	Symbol	Description	Reset Value
7:0	RBR	The UARTn Receiver Buffer Register contains the oldest received byte in the UARTn Rx FIFO.	Undefined
31:8	-	Reserved, the value read from a reserved bit is not defined.	NA

4.2 UARTn Transmit Holding Register (U0THR - 0x4000 C000, U2THR - 0x4009 8000, U3THR - 0x4009 C000 when DLAB = 0)

The UnTHR is the top byte of the UARTn TX FIFO. The top byte is the newest character in the TX FIFO and can be written via the bus interface. The LSB represents the first bit to transmit.

The Divisor Latch Access Bit (DLAB) in UnLCR must be zero in order to access the UnTHR. The UnTHR is always write-only.

Table 271: UARTn Transmit Holding Register (U0THR - address 0x4000 C000, U2THR - 0x4009 8000, U3THR - 0x4009 C000 when DLAB = 0) bit description

Bit	Symbol	Description	Reset Value
7:0	THR	Writing to the UARTn Transmit Holding Register causes the data to be stored in the UARTn transmit FIFO. The byte will be sent when it reaches the bottom of the FIFO and the transmitter is available.	NA
31:8	-	Reserved, user software should not write ones to reserved bits.	NA

4.3 UARTn Divisor Latch LSB register (U0DLL - 0x4000 C000, U2DLL - 0x4009 8000, U3DLL - 0x4009 C000 when DLAB = 1) and UARTn Divisor Latch MSB register (U0DLM - 0x4000 C004, U2DLL - 0x4009 8004, U3DLL - 0x4009 C004 when DLAB = 1)

The UARTn Divisor Latch is part of the UARTn Baud Rate Generator and holds the value used, along with the Fractional Divider, to divide the APB clock (PCLK) in order to produce the baud rate clock, which must be 16× the desired baud rate. The UnDLL and UnDLM registers together form a 16-bit divisor where UnDLL contains the lower 8 bits of the divisor and UnDLM contains the higher 8 bits of the divisor. A 0x0000 value is treated like a 0x0001 value as division by zero is not allowed. The Divisor Latch Access Bit (DLAB) in

UnLCR must be one in order to access the UARTn Divisor Latches. Details on how to select the right value for U1DLL and U1DLM can be found later in this chapter, see Section 14–4.12.

Table 272: UARTn Divisor Latch LSB register (U0DLL - address 0x4000 C000, U2DLL - 0x4009 8000, U3DLL - 0x4009 C000 when DLAB = 1) bit description

Bit	Symbol	Description	Reset Value
7:0	DLLSB	The UARTn Divisor Latch LSB Register, along with the UnDLM register, determines the baud rate of the UARTn.	0x01
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 273: UARTn Divisor Latch MSB register (U0DLM - address 0x4000 C004, U2DLM - 0x4009 8004, U3DLM - 0x4009 C004 when DLAB = 1) bit description

Bit	Symbol	Description	Reset Value
7:0	DLMSB	The UARTn Divisor Latch MSB Register, along with the U0DLL register, determines the baud rate of the UARTn.	0x00
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

4.4 UARTn Interrupt Enable Register (U0IER - 0x4000 C004, U2IER - 0x4009 8004, U3IER - 0x4009 C004 when DLAB = 0)

The UnIER is used to enable the three UARTn interrupt sources.

Table 274: UARTn Interrupt Enable Register (U0IER - address 0x4000 C004, U2IER - 0x4009 8004, U3IER - 0x4009 C004 when DLAB = 0) bit description

Bit	Symbol	Value	Description	Reset Value
0	RBR Interrupt Enable		Enables the Receive Data Available interrupt for UARTn. It also controls the Character Receive Time-out interrupt.	0
		0	Disable the RDA interrupts.	
		1	Enable the RDA interrupts.	
1	THRE Interrupt Enable		Enables the THRE interrupt for UARTn. The status of this can be read from UnLSR[5].	0
		0	Disable the THRE interrupts.	
		1	Enable the THRE interrupts.	
2	RX Line Status Interrupt Enable		Enables the UARTn RX line status interrupts. The status of this interrupt can be read from UnLSR[4:1].	0
		0	Disable the RX line status interrupts.	
		1	Enable the RX line status interrupts.	
7:3	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
8	ABEOIntEn		Enables the end of auto-baud interrupt.	0
		0	Disable end of auto-baud Interrupt.	
		1	Enable end of auto-baud Interrupt.	

Table 274: UARTn Interrupt Enable Register (U0IER - address 0x4000 C004, U2IER - 0x4009 8004, U3IER - 0x4009 C004 when DLAB = 0) bit description

Bit	Symbol	Value	Description	Reset Value
9	ABTOIntEn		Enables the auto-baud time-out interrupt.	0
		0	Disable auto-baud time-out Interrupt.	
		1	Enable auto-baud time-out Interrupt.	
31:10	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

4.5 UARTn Interrupt Identification Register (U0IIR - 0x4000 C008, U2IIR - 0x4009 8008, U3IIR - 0x4009 C008)

The UnIIR provides a status code that denotes the priority and source of a pending interrupt. The interrupts are frozen during an UnIIR access. If an interrupt occurs during an UnIIR access, the interrupt is recorded for the next UnIIR access.

Table 275: UARTn Interrupt Identification Register (U0IIR - address 0x4000 C008, U2IIR - 0x4009 8008, U3IIR - 0x4009 C008) bit description

Bit	Symbol	Value	Description	Reset Value
0	IntStatus		Interrupt status. Note that U1IIR[0] is active low. The pending interrupt can be determined by evaluating UnIIR[3:1].	1
		0	At least one interrupt is pending.	
		1	No interrupt is pending.	
3:1	IntId		Interrupt identification. UnIER[3:1] identifies an interrupt corresponding to the UARTn Rx or TX FIFO. All other combinations of UnIER[3:1] not listed below are reserved (000,100,101,111).	0
		011	1 - Receive Line Status (RLS).	
		010	2a - Receive Data Available (RDA).	
		110	2b - Character Time-out Indicator (CTI).	
		001	3 - THRE Interrupt	
5:4	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7:6	FIFO Enable		Copies of UnFCR[0].	0
8	ABEOInt		End of auto-baud interrupt. True if auto-baud has finished successfully and interrupt is enabled.	0
9	ABTOInt		Auto-baud time-out interrupt. True if auto-baud has timed out and interrupt is enabled.	0
31:10	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Bit UnIIR[9:8] are set by the auto-baud function and signal a time-out or end of auto-baud condition. The auto-baud interrupt conditions are cleared by setting the corresponding Clear bits in the Auto-baud Control Register.

If the IntStatus bit is 1 no interrupt is pending and the IntId bits will be zero. If the IntStatus is 0, a non auto-baud interrupt is pending in which case the IntId bits identify the type of interrupt and handling as described in Table 14–276. Given the status of UnIIR[3:0], an interrupt handler routine can determine the cause of the interrupt and how to clear the active interrupt. The UnIIR must be read in order to clear the interrupt prior to exiting the Interrupt Service Routine.

The UARTn RLS interrupt (UnIIR[3:1] = 011) is the highest priority interrupt and is set whenever any one of four error conditions occur on the UARTn Rx input: overrun error (OE), parity error (PE), framing error (FE) and break interrupt (BI). The UARTn Rx error condition that set the interrupt can be observed via U0LSR[4:1]. The interrupt is cleared upon an UnLSR read.

The UARTn RDA interrupt (UnIIR[3:1] = 010) shares the second level priority with the CTI interrupt (UnIIR[3:1] = 110). The RDA is activated when the UARTn Rx FIFO reaches the trigger level defined in UnFCR[7:6] and is reset when the UARTn Rx FIFO depth falls below the trigger level. When the RDA interrupt goes active, the CPU can read a block of data defined by the trigger level.

The CTI interrupt (UnIIR[3:1] = 110) is a second level interrupt and is set when the UARTn Rx FIFO contains at least one character and no UARTn Rx FIFO activity has occurred in 3.5 to 4.5 character times. Any UARTn Rx FIFO activity (read or write of UARTn RSR) will clear the interrupt. This interrupt is intended to flush the UARTn RBR after a message has been received that is not a multiple of the trigger level size. For example, if a peripheral wished to send a 105 character message and the trigger level was 10 characters, the CPU would receive 10 RDA interrupts resulting in the transfer of 100 characters and 1 to 5 CTI interrupts (depending on the service routine) resulting in the transfer of the remaining 5 characters.

Table 276: UARTn Interrupt Handling

U0IIR[3:0] value[1]	Priority	Interrupt Type	Interrupt Source	Interrupt Reset
0001	-	None	None	-
0110	Highest	RX Line Status / Error	OE[2] or PE[2] or FE[2] or BI[2]	UnLSR Read[2]
0100	Second	RX Data Available	Rx data available or trigger level reached in FIFO (UnFCR0=1)	UnRBR Read[3] or UARTn FIFO drops below trigger level
1100	Second	Character Time-out indication	Minimum of one character in the Rx FIFO and no character input or removed during a time period depending on how many characters are in FIFO and what the trigger level is set at (3.5 to 4.5 character times). The exact time will be: [(word length) $\times$ 7 - 2] $\times$ 8 + [(trigger level - number of characters) $\times$ 8 + 1] RCLKs	UnRBR Read[3]
0010	Third	THRE	THRE[2]	UnIIR Read (if source of interrupt) or THR write[4]

[1] Values "0000", “0011”, “0101”, “0111”, “1000”, “1001”, “1010”, “1011”,”1101”,”1110”,”1111” are reserved.

[2] For details see Section 14–4.8 “UARTn Line Status Register (U0LSR - 0x4000 C014, U2LSR - 0x4009 8014, U3LSR - 0x4009 C014)”

[3] For details see Section 14–14.4.1 “UARTn Receiver Buffer Register (U0RBR - 0x4000 C000, U2RBR - 0x4009 8000, U3RBR - 0x4009 C000 when DLAB = 0)”

[4] For details see Section 14–4.5 “UARTn Interrupt Identification Register (U0IIR - 0x4000 C008, U2IIR - 0x4009 8008, U3IIR - 0x4009 C008)” and Section 14–4.2 “UARTn Transmit Holding Register (U0THR - 0x4000 C000, U2THR - 0x4009 8000, U3THR - 0x4009 C000 when DLAB = 0)”

The UARTn THRE interrupt (UnIIR[3:1] = 001) is a third level interrupt and is activated when the UARTn THR FIFO is empty provided certain initialization conditions have been met. These initialization conditions are intended to give the UARTn THR FIFO a chance to fill up with data to eliminate many THRE interrupts from occurring at system start-up. The initialization conditions implement a one character delay minus the stop bit whenever THRE = 1 and there have not been at least two characters in the UnTHR at one time since the last THRE = 1 event. This delay is provided to give the CPU time to write data to UnTHR without a THRE interrupt to decode and service. A THRE interrupt is set immediately if the UARTn THR FIFO has held two or more characters at one time and currently, the UnTHR is empty. The THRE interrupt is reset when a UnTHR write occurs or a read of the UnIIR occurs and the THRE is the highest interrupt (UnIIR[3:1] = 001).

4.6 UARTn FIFO Control Register (U0FCR - 0x4000 C008, U2FCR - 0x4009 8008, U3FCR - 0x4009 C008)

The write-only UnFCR controls the operation of the UARTn Rx and TX FIFOs.

Table 277: UARTn FIFO Control Register (U0FCR - address 0x4000 C008, U2FCR - 0x4009 8008, U3FCR - 0x4007 C008) bit description

Bit	Symbol	Value	Description	Reset Value
0	FIFO Enable	0	UARTn FIFOs are disabled. Must not be used in the application.	0
		1	Active high enable for both UARTn Rx and TX FIFOs and UnFCR[7:1] access. This bit must be set for proper UART operation. Any transition on this bit will automatically clear the related UART FIFOs.	
1	RX FIFO Reset	0	No impact on either of UARTn FIFOs.	0
		1	Writing a logic 1 to UnFCR[1] will clear all bytes in UARTn Rx FIFO, reset the pointer logic. This bit is self-clearing.	
2	TX FIFO Reset	0	No impact on either of UARTn FIFOs.	0
		1	Writing a logic 1 to UnFCR[2] will clear all bytes in UARTn TX FIFO, reset the pointer logic. This bit is self-clearing.	
3	DMA Mode Select		When the FIFO enable bit (bit 0 of this register) is set, this bit selects the DMA mode. See Section 14–4.6.1.	0
5:4	-	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7:6	RX Trigger Level		These two bits determine how many receiver UARTn FIFO characters must be written before an interrupt or DMA request is activated.	0
		00	Trigger level 0 (1 character or 0x01)	
		01	Trigger level 1 (4 characters or 0x04)	
		10	Trigger level 2 (8 characters or 0x08)	
		11	Trigger level 3 (14 characters or 0x0E)	
31:8	-		Reserved, user software should not write ones to reserved bits.	NA

4.6.1 DMA Operation

The user can optionally operate the UART transmit and/or receive using DMA. The DMA mode is determined by the DMA Mode Select bit in the FCR register. This bit only has an affect when the FIFOs are enabled via the FIFO Enable bit in the FCR register.

UART receiver DMA

In DMA mode, the receiver DMA request is asserted on the event of the receiver FIFO level becoming equal to or greater than trigger level, or if a character timeout occurs. See the description of the RX Trigger Level above. The receiver DMA request is cleared by the DMA controller.

UART transmitter DMA

In DMA mode, the transmitter DMA request is asserted on the event of the transmitter FIFO transitioning to not full. The transmitter DMA request is cleared by the DMA controller.

4.7 UARTn Line Control Register (U0LCR - 0x4000 C00C, U2LCR - 0x4009 800C, U3LCR - 0x4009 C00C)

The UnLCR determines the format of the data character that is to be transmitted or received.

Table 278: UARTn Line Control Register (U0LCR - address 0x4000 C00C, U2LCR - 0x4009 800C, U3LCR - 0x4009 C00C) bit description

Bit	Symbol	Value	Description	Reset Value
1:0	Word Length Select	00	5-bit character length	0
		01	6-bit character length	
		10	7-bit character length	
		11	8-bit character length	
2	Stop Bit Select	0	1 stop bit.	0
		1	2 stop bits (1.5 if UnLCR[1:0]=00).	
3	Parity Enable	0	Disable parity generation and checking.	0
		1	Enable parity generation and checking.	
5:4	Parity Select	00	Odd parity. Number of 1s in the transmitted character and the attached parity bit will be odd.	0
		01	Even Parity. Number of 1s in the transmitted character and the attached parity bit will be even.	
		10	Forced "1" stick parity.	
		11	Forced "0" stick parity.	
6	Break Control	0	Disable break transmission.	0
		1	Enable break transmission. Output pin UARTn TXD is forced to logic 0 when UnLCR[6] is active high.	
7	Divisor Latch Access Bit (DLAB)	0	Disable access to Divisor Latches.	0
		1	Enable access to Divisor Latches.	
31:8	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

4.8 UARTn Line Status Register (U0LSR - 0x4000 C014, U2LSR - 0x4009 8014, U3LSR - 0x4009 C014)

The UnLSR is a read-only register that provides status information on the UARTn TX and RX blocks.

Table 279: UARTn Line Status Register (U0LSR - address 0x4000 C014, U2LSR - 0x4009 8014, U3LSR - 0x4009 C014) bit description

Bit	Symbol	Value	Description	Reset Value
0	Receiver Data Ready (RDR)		UnLSR0 is set when the UnRBR holds an unread character and is cleared when the UARTn RBR FIFO is empty.	0
		0	The UARTn receiver FIFO is empty.	
		1	The UARTn receiver FIFO is not empty.	
1	Overrun Error (OE)		The overrun error condition is set as soon as it occurs. An UnLSR read clears UnLSR1. UnLSR1 is set when UARTn RSR has a new character assembled and the UARTn RBR FIFO is full. In this case, the UARTn RBR FIFO will not be overwritten and the character in the UARTn RSR will be lost.	0
		0	Overrun error status is inactive.	
		1	Overrun error status is active.	
2	Parity Error (PE)		When the parity bit of a received character is in the wrong state, a parity error occurs. An UnLSR read clears UnLSR[2]. Time of parity error detection is dependent on UnFCR[0]. **Note:** A parity error is associated with the character at the top of the UARTn RBR FIFO.	0
		0	Parity error status is inactive.	
		1	Parity error status is active.	
3	Framing Error (FE)		When the stop bit of a received character is a logic 0, a framing error occurs. An UnLSR read clears UnLSR[3]. The time of the framing error detection is dependent on UnFCR0. Upon detection of a framing error, the Rx will attempt to resynchronize to the data and assume that the bad stop bit is actually an early start bit. However, it cannot be assumed that the next received byte will be correct even if there is no Framing Error. **Note:** A framing error is associated with the character at the top of the UARTn RBR FIFO.	0
		0	Framing error status is inactive.	
		1	Framing error status is active.	
4	Break Interrupt (BI)		When RXDn is held in the spacing state (all zeroes) for one full character transmission (start, data, parity, stop), a break interrupt occurs. Once the break condition has been detected, the receiver goes idle until RXDn goes to marking state (all ones). An UnLSR read clears this status bit. The time of break detection is dependent on UnFCR[0]. **Note:** The break interrupt is associated with the character at the top of the UARTn RBR FIFO.	0
		0	Break interrupt status is inactive.	
		1	Break interrupt status is active.	
5	Transmitter Holding Register Empty (THRE))		THRE is set immediately upon detection of an empty UARTn THR and is cleared on a UnTHR write.	1
		0	UnTHR contains valid data.	
		1	UnTHR is empty.	
6	Transmitter Empty (TEMT)		TEMT is set when both UnTHR and UnTSR are empty; TEMT is cleared when either the UnTSR or the UnTHR contain valid data.	1
		0	UnTHR and/or the UnTSR contains valid data.	
		1	UnTHR and the UnTSR are empty.	

Table 279: UARTn Line Status Register (U0LSR - address 0x4000 C014, U2LSR - 0x4009 8014, U3LSR - 0x4009 C014) bit description

Bit	Symbol	Value	Description	Reset Value
7	Error in RX FIFO (RXFE)		UnLSR[7] is set when a character with a Rx error such as framing error, parity error or break interrupt, is loaded into the UnRBR. This bit is cleared when the UnLSR register is read and there are no subsequent errors in the UARTn FIFO.	0
		0	UnRBR contains no UARTn RX errors or UnFCR[0]=0.	
		1	UARTn RBR contains at least one UARTn RX error.	
31:8	-		Reserved, the value read from a reserved bit is not defined.	NA

4.9 UARTn Scratch Pad Register (U0SCR - 0x4000 C01C, U2SCR - 0x4009 801C U3SCR - 0x4009 C01C)

The UnSCR has no effect on the UARTn operation. This register can be written and/or read at user's discretion. There is no provision in the interrupt interface that would indicate to the host that a read or write of the UnSCR has occurred.

Table 280: UARTn Scratch Pad Register (U0SCR - address 0x4000 C01C, U2SCR - 0x4009 801C, U3SCR - 0x4009 C01C) bit description

Bit	Symbol	Description	Reset Value
7:0	Pad	A readable, writable byte.	0x00
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

4.10 UARTn Auto-baud Control Register (U0ACR - 0x4000 C020, U2ACR - 0x4009 8020, U3ACR - 0x4009 C020)

The UARTn Auto-baud Control Register (UnACR) controls the process of measuring the incoming clock/data rate for the baud rate generation and can be read and written at user's discretion.

Table 281: UARTn Auto-baud Control Register (U0ACR - address 0x4000 C020, U2ACR - 0x4009 8020, U3ACR - 0x4009 C020) bit description

Bit	Symbol	Value	Description	Reset value
0	Start		This bit is automatically cleared after auto-baud completion.	0
		0	Auto-baud stop (auto-baud is not running).	
		1	Auto-baud start (auto-baud is running). Auto-baud run bit. This bit is automatically cleared after auto-baud completion.	
1	Mode		Auto-baud mode select bit.	0
		0	Mode 0.	
		1	Mode 1.	
2	AutoRestart	0	No restart.	0
		1	Restart in case of time-out (counter restarts at next UARTn Rx falling edge)	0
7:3	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 281: UARTn Auto-baud Control Register (U0ACR - address 0x4000 C020, U2ACR - 0x4009 8020, U3ACR - 0x4009 C020) bit description

Bit	Symbol	Value	Description	Reset value
8	ABEOIntClr		End of auto-baud interrupt clear bit (write-only accessible). Writing a 1 will clear the corresponding interrupt in the UnIIR. Writing a 0 has no impact.	0
9	ABTOIntClr		Auto-baud time-out interrupt clear bit (write-only accessible). Writing a 1 will clear the corresponding interrupt in the UnIIR. Writing a 0 has no impact.	0
31:10	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

14.4.10.1 Auto-baud

The UARTn auto-baud function can be used to measure the incoming baud-rate based on the “AT” protocol (Hayes command). If enabled the auto-baud feature will measure the bit time of the receive data stream and set the divisor latch registers UnDLM and UnDLL accordingly.

Remark: the fractional rate divider is not connected during auto-baud operations, and therefore should not be used when the auto-baud feature is needed.

Auto-baud is started by setting the UnACR Start bit. Auto-baud can be stopped by clearing the UnACR Start bit. The Start bit will clear once auto-baud has finished and reading the bit will return the status of auto-baud (pending/finished).

Two auto-baud measuring modes are available which can be selected by the UnACR Mode bit. In mode 0 the baud-rate is measured on two subsequent falling edges of the UARTn Rx pin (the falling edge of the start bit and the falling edge of the least significant bit). In mode 1 the baud-rate is measured between the falling edge and the subsequent rising edge of the UARTn Rx pin (the length of the start bit).

The UnACR AutoRestart bit can be used to automatically restart baud-rate measurement if a time-out occurs (the rate measurement counter overflows). If this bit is set the rate measurement will restart at the next falling edge of the UARTn Rx pin.

The auto-baud function can generate two interrupts.

- The UnIIR ABTOInt interrupt will get set if the interrupt is enabled (UnIER ABToIntEn is set and the auto-baud rate measurement counter overflows).
- The UnIIR ABEOInt interrupt will get set if the interrupt is enabled (UnIER ABEOIntEn is set and the auto-baud has completed successfully).

The auto-baud interrupts have to be cleared by setting the corresponding UnACR ABTOIntClr and ABEOIntEn bits.

Typically the fractional baud-rate generator is disabled (DIVADDVAL = 0) during auto-baud. However, if the fractional baud-rate generator is enabled (DIVADDVAL > 0), it is going to impact the measuring of UARTn Rx pin baud-rate, but the value of the UnFDR register is not going to be modified after rate measurement. Also, when auto-baud is used, any write to UnDLM and UnDLL registers should be done before UnACR register write. The minimum and the maximum baud rates supported by UARTn are function of pclk, number of data bits, stop bits and parity bits.

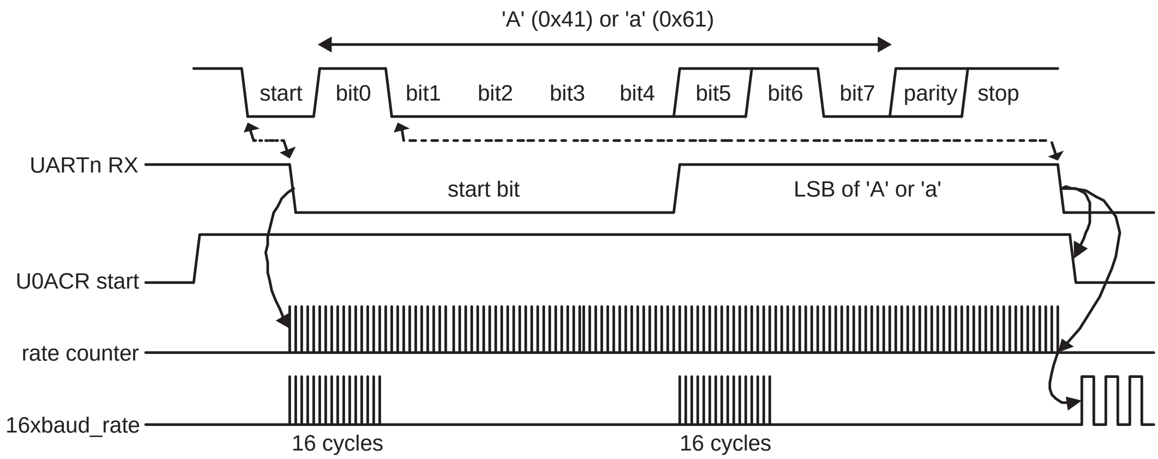
(1)

$$ratemin = \frac{2 \times PCLK}{16 \times 2^{15}} \leq UARTn_{baudrate} \leq \frac{PCLK}{16 \times (2 + databits + paritybits + stopbits)} = ratemax$$

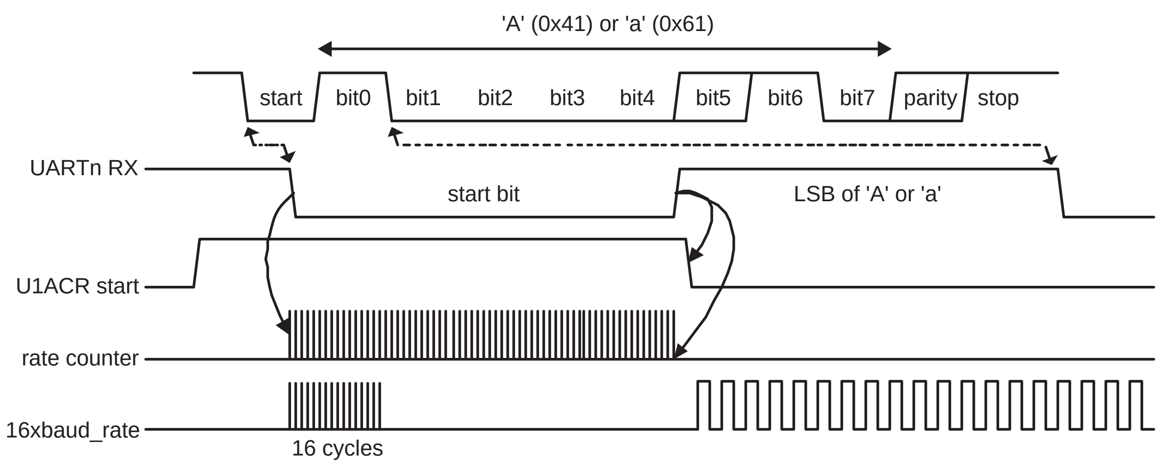
14.4.10.2 Auto-baud modes

When the software is expecting an “AT” command, it configures the UARTn with the expected character format and sets the UnACR Start bit. The initial values in the divisor latches UnDLM and UnDLM don‘t care. Because of the “A” or “a” ASCII coding (”A" = 0x41, “a” = 0x61), the UARTn Rx pin sensed start bit and the LSB of the expected character are delimited by two falling edges. When the UnACR Start bit is set, the auto-baud protocol will execute the following phases:

1. On UnACR Start bit setting, the baud rate measurement counter is reset and the UARTn UnRSR is reset. The UnRSR baud rate is switch to the highest rate.
2. A falling edge on UARTn Rx pin triggers the beginning of the start bit. The rate measuring counter will start counting pclk cycles optionally pre-scaled by the fractional baud-rate generator.
3. During the receipt of the start bit, 16 pulses are generated on the RSR baud input with the frequency of the (fractional baud-rate pre-scaled) UARTn input clock, guaranteeing the start bit is stored in the UnRSR.
4. During the receipt of the start bit (and the character LSB for mode = 0) the rate counter will continue incrementing with the pre-scaled UARTn input clock (pclk).
5. If Mode = 0 then the rate counter will stop on next falling edge of the UARTn Rx pin. If Mode = 1 then the rate counter will stop on the next rising edge of the UARTn Rx pin.
6. The rate counter is loaded into UnDLM/UnDLL and the baud-rate will be switched to normal operation. After setting the UnDLM/UnDLL the end of auto-baud interrupt UnIIR ABEOInt will be set, if enabled. The UnRSR will now continue receiving the remaining bits of the “A/a” character.



a. Mode 0 (start bit and LSB are used for auto-baud)



b. Mode 1 (only start bit is used for auto-baud)

Fig 44. Auto-baud a) mode 0 and b) mode 1 waveform

4.11 UARTn IrDA Control Register (U0ICR - 0x4000 C024, U2ICR - 0x4009 8024, U3ICR - 0x4009 C024)

The IrDA Control Register enables and configures the IrDA mode on each UART. The value of UnICR should not be changed while transmitting or receiving data, or data loss or corruption may occur.

Table 282: UARTn IrDA Control Register (U0ICR - 0x4000 C024, U2ICR - 0x4009 8024, U3ICR - 0x4009 C024) bit description

Bit	Symbol	Value	Description	Reset value
0	IrDAEn	0	IrDA mode on UARTn is disabled, UARTn acts as a standard UART.	0
		1	IrDA mode on UARTn is enabled.	
1	IrDAInv		When 1, the serial input is inverted. This has no effect on the serial output. When 0, the serial input is not inverted.	0

Table 282: UARTn IrDA Control Register (U0ICR - 0x4000 C024, U2ICR - 0x4009 8024, U3ICR - 0x4009 C024) bit description

Bit	Symbol	Value	Description	Reset value
2	FixPulseEn		When 1, enabled IrDA fixed pulse width mode.	0
5:3	PulseDiv		Configures the pulse when FixPulseEn = 1. See text below for details.	0
31:6	-	NA	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	0

The PulseDiv bits in UnICR are used to select the pulse width when the fixed pulse width mode is used in IrDA mode (IrDAEn = 1 and FixPulseEn = 1). The value of these bits should be set so that the resulting pulse width is at least 1.63 μs. Table 14–283 shows the possible pulse widths.

Table 283: IrDA Pulse Width

FixPulseEn	PulseDiv	IrDA Transmitter Pulse width (μs)
0	x	$3 / (16 \times$ baud rate)
1	0	$2 \times T_{PCLK}$
1	1	$4 \times T_{PCLK}$
1	2	$8 \times T_{PCLK}$
1	3	$16 \times T_{PCLK}$
1	4	$32 \times T_{PCLK}$
1	5	$64 \times T_{PCLK}$
1	6	$128 \times T_{PCLK}$
1	7	$256 \times T_{PCLK}$

4.12 UARTn Fractional Divider Register (U0FDR - 0x4000 C028, U2FDR - 0x4009 8028, U3FDR - 0x4009 C028)

The UART0/2/3 Fractional Divider Register (U0/2/3FDR) controls the clock pre-scaler for the baud rate generation and can be read and written at the user's discretion. This pre-scaler takes the APB clock and generates an output clock according to the specified fractional requirements.

Important: If the fractional divider is active (DIVADDVAL > 0) and DLM = 0, the value of the DLL register must be greater than 2.

Table 284: UARTn Fractional Divider Register (U0FDR - address 0x4000 C028, U2FDR - 0x4009 8028, U3FDR - 0x4009 C028) bit description

Bit	Function	Value	Description	Reset value
3:0	DIVADDVAL	0	Baud-rate generation pre-scaler divisor value. If this field is 0, fractional baud-rate generator will not impact the UARTn baudrate.	0
7:4	MULVAL	1	Baud-rate pre-scaler multiplier value. This field must be greater or equal 1 for UARTn to operate properly, regardless of whether the fractional baud-rate generator is used or not.	1
31:8	-	NA	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	0

This register controls the clock pre-scaler for the baud rate generation. The reset value of the register keeps the fractional capabilities of UART0/2/3 disabled making sure that UART0/2/3 is fully software and hardware compatible with UARTs not equipped with this feature.

UART0/2/3 baud rate can be calculated as (n = 0/2/3):

$$UARTn_{baudrate} = \frac{PCLK}{16 \times (256 \times UnDLM + UnDLL) \times \left(1 + \frac{DivAddVal}{MulVal}\right)} \tag{2}$$

Where PCLK is the peripheral clock, U0/2/3DLM and U0/2/3DLL are the standard UART0/2/3 baud rate divider registers, and DIVADDVAL and MULVAL are UART0/2/3 fractional baud rate generator specific parameters.

The value of MULVAL and DIVADDVAL should comply to the following conditions:

1. $1 \leq$ MULVAL ≤ 15
2. $0 \leq$ DIVADDVAL ≤ 14
3. DIVADDVAL < MULVAL

The value of the U0/2/3FDR should not be modified while transmitting/receiving data or data may be lost or corrupted.

If the U0/2/3FDR register value does not comply to these two requests, then the fractional divider output is undefined. If DIVADDVAL is zero then the fractional divider is disabled, and the clock will not be divided.

4.12.1 Baud rate calculation

UARTn can operate with or without using the Fractional Divider. In real-life applications it is likely that the desired baud rate can be achieved using several different Fractional Divider settings. The following algorithm illustrates one way of finding a set of DLM, DLL, MULVAL, and DIVADDVAL values. Such set of parameters yields a baud rate with a relative error of less than 1.1% from the desired one.

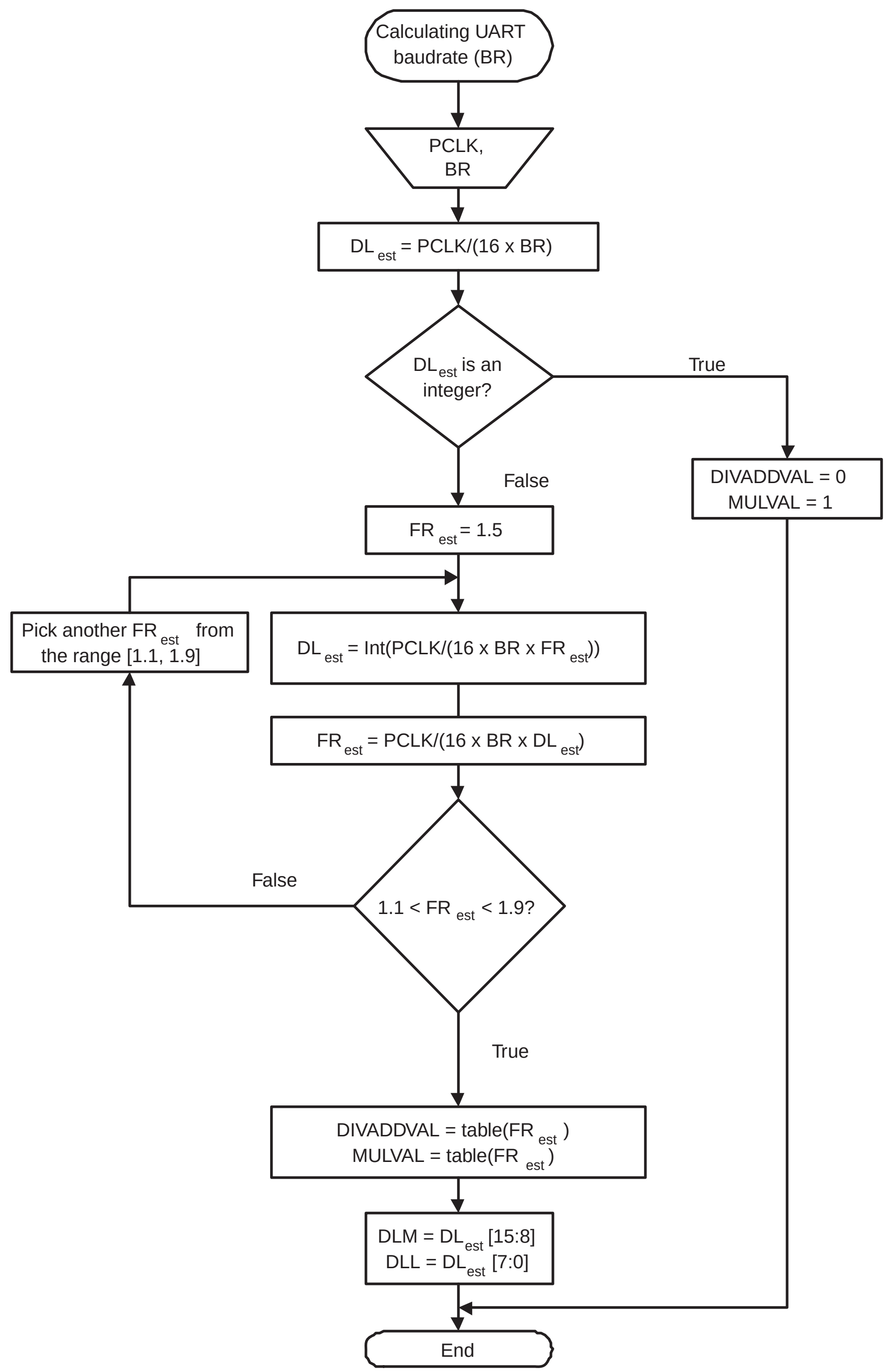


Fig 45. Algorithm for setting UART dividers

Table 285. Fractional Divider setting look-up table

FR	DivAddVal/ MulVal	FR	DivAddVal/ MulVal	FR	DivAddVal/ MulVal	FR	DivAddVal/ MulVal
1.000	0/1	1.250	1/4	1.500	1/2	1.750	3/4
1.067	1/15	1.267	4/15	1.533	8/15	1.769	10/13
1.071	1/14	1.273	3/11	1.538	7/13	1.778	7/9
1.077	1/13	1.286	2/7	1.545	6/11	1.786	11/14
1.083	1/12	1.300	3/10	1.556	5/9	1.800	4/5
1.091	1/11	1.308	4/13	1.571	4/7	1.818	9/11
1.100	1/10	1.333	1/3	1.583	7/12	1.833	5/6
1.111	1/9	1.357	5/14	1.600	3/5	1.846	11/13
1.125	1/8	1.364	4/11	1.615	8/13	1.857	6/7
1.133	2/15	1.375	3/8	1.625	5/8	1.867	13/15
1.143	1/7	1.385	5/13	1.636	7/11	1.875	7/8
1.154	2/13	1.400	2/5	1.643	9/14	1.889	8/9
1.167	1/6	1.417	5/12	1.667	2/3	1.900	9/10
1.182	2/11	1.429	3/7	1.692	9/13	1.909	10/11
1.200	1/5	1.444	4/9	1.700	7/10	1.917	11/12
1.214	3/14	1.455	5/11	1.714	5/7	1.923	12/13
1.222	2/9	1.462	6/13	1.727	8/11	1.929	13/14
1.231	3/13	1.467	7/15	1.733	11/15	1.933	14/15

4.12.1.1 Example 1: PCLK = 14.7456 MHz, BR = 9600

According to the provided algorithm DL_{est} = PCLK/(16 x BR) = 14.7456 MHz / (16 x 9600) = 96. Since this DL_{est} is an integer number, DIVADDVAL = 0, MULVAL = 1, DLM = 0, and DLL = 96.

4.12.1.2 Example 2: PCLK = 12 MHz, BR = 115200

According to the provided algorithm DL_{est} = PCLK/(16 x BR) = 12 MHz / (16 x 115200) = 6.51. This DL_{est} is not an integer number and the next step is to estimate the FR parameter. Using an initial estimate of FR_{est} = 1.5 a new DL_{est} = 4 is calculated and FR_{est} is recalculated as FR_{est} = 1.628. Since FRest = 1.628 is within the specified range of 1.1 and 1.9, DIVADDVAL and MULVAL values can be obtained from the attached look-up table.

The closest value for FRest = 1.628 in the look-up Table 14–285 is FR = 1.625. It is equivalent to DIVADDVAL = 5 and MULVAL = 8.

Based on these findings, the suggested UART setup would be: DLM = 0, DLL = 4, DIVADDVAL = 5, and MULVAL = 8. According to Equation 14–2 the UART rate is 115384. This rate has a relative error of 0.16% from the originally specified 115200.

4.13 UARTn Transmit Enable Register (U0TER - 0x4000 C030, U2TER - 0x4009 8030, U3TER - 0x4009 C030)

The UnTER register enables implementation of software flow control. When TXEn=1, UARTn transmitter will keep sending data as long as they are available. As soon as TXEn becomes 0, UARTn transmission will stop.

Table 14–286 describes how to use TXEn bit in order to achieve software flow control.

Table 286: UARTn Transmit Enable Register (U0TER - address 0x4000 C030, U2TER - 0x4009 8030, U3TER - 0x4009 C030) bit description

Bit	Symbol	Description	Reset Value
6:0	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7	TXEN	When this bit is 1, as it is after a Reset, data written to the THR is output on the TXD pin as soon as any preceding data has been sent. If this bit is cleared to 0 while a character is being sent, the transmission of that character is completed, but no further characters are sent until this bit is set again. In other words, a 0 in this bit blocks the transfer of characters from the THR or TX FIFO into the transmit shift register. Software implementing software-handshaking can clear this bit when it receives an XOFF character (DC3). Software can set this bit again when it receives an XON (DC1) character.	1

4.14 UARTn FIFO Level register (U0FIFOLVL - 0x4000 C058, U2FIFOLVL - 0x4009 8058, U3FIFOLVL - 0x4009 C058)

UnFIFOLVL register is a read-only register that allows software to read the current FIFO level status. Both the transmit and receive FIFO levels are present in this register.

Table 287. UARTn FIFO Level register (U0FIFOLVL - 0x4000 C058, U2FIFOLVL - 0x4009 8058, U3FIFOLVL - 0x4009 C058) bit description

Bit	Symbol	Description	Reset value
3:0	RXFIFILVL	Reflects the current level of the UART receiver FIFO. 0 = empty, 0xF = FIFO full.	0x00
7:4	-	Reserved. The value read from a reserved bit is not defined.	NA
11:8	TXFIFOLVL	Reflects the current level of the UART transmitter FIFO. 0 = empty, 0xF = FIFO full.	0x00
31:12	-	Reserved. The value read from a reserved bit is not defined.	NA

5. Architecture

The architecture of the UARTs 0, 2 and 3 are shown below in the block diagram.

The APB interface provides a communications link between the CPU or host and the UART.

The UARTn receiver block, UnRX, monitors the serial input line, RXDn, for valid input. The UARTn RX Shift Register (UnRSR) accepts valid characters via RXDn. After a valid character is assembled in the UnRSR, it is passed to the UARTn RX Buffer Register FIFO to await access by the CPU or host via the generic host interface.

The UARTn transmitter block, UnTX, accepts data written by the CPU or host and buffers the data in the UARTn TX Holding Register FIFO (UnTHR). The UARTn TX Shift Register (UnTSR) reads the data stored in the UnTHR and assembles the data to transmit via the serial output pin, TXDn.

The UARTn Baud Rate Generator block, UnBRG, generates the timing enables used by the UARTn TX block. The UnBRG clock input source is the APB clock (PCLK). The main clock is divided down per the divisor specified in the UnDLL and UnDLM registers. This divided down clock is a 16x oversample clock, NBAUDOUT.

The interrupt interface contains registers UnIER and UnIIR. The interrupt interface receives several one clock wide enables from the UnTX and UnRX blocks.

Status information from the UnTX and UnRX is stored in the UnLSR. Control information for the UnTX and UnRX is stored in the UnLCR.

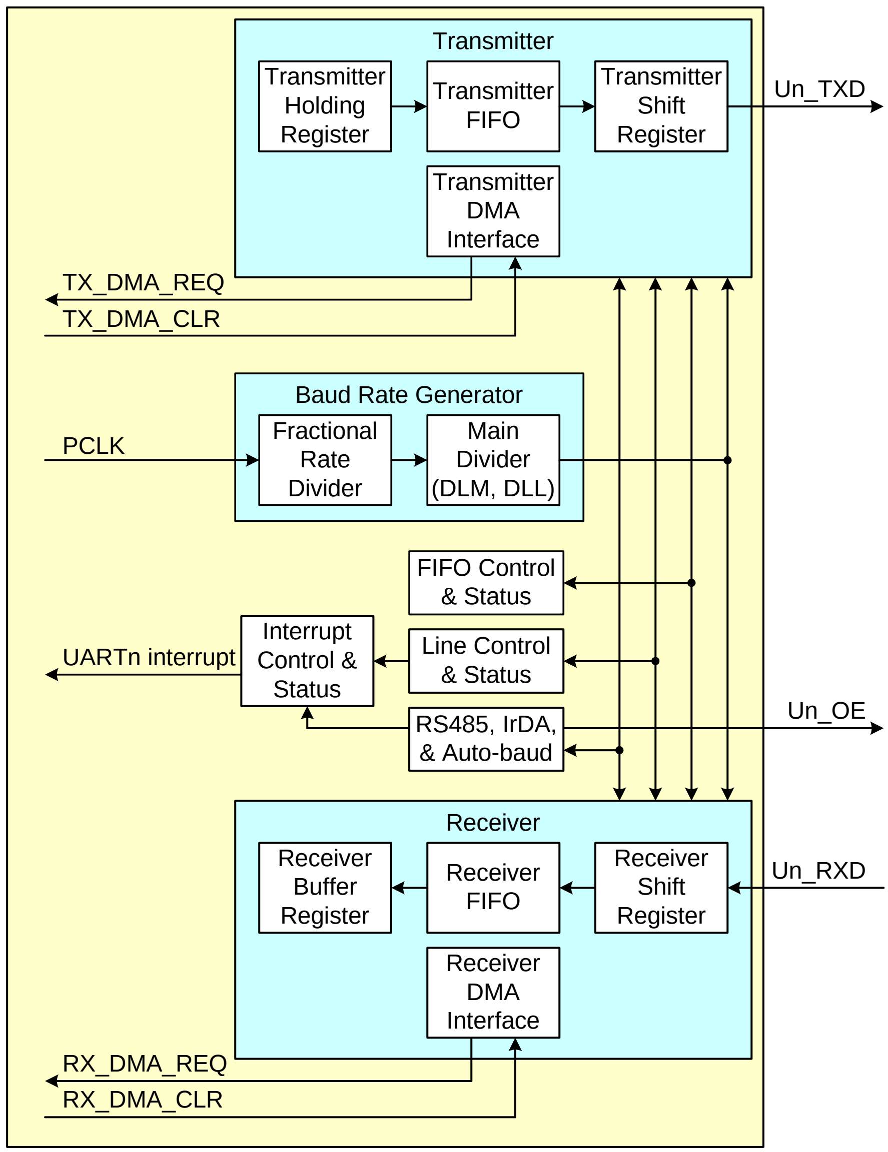


Fig 46. UART0, 2 and 3 block diagram

UM10360

Chapter 15: LPC17xx UART1

Rev. 01 — 4 January 2010 User manual

1. Basic configuration

The UART1 peripheral is configured using the following registers:

1. Power: In the PCONP register (Table 4–46), set bits PCUART1.

   Remark: On reset, UART1 is enabled (PCUART1 = 1).
2. Peripheral clock: In the PCLKSEL0 register (Table 4–40), select PCLK_UART1.
3. Baud rate: In register U1LCR (Table 15–298), set bit DLAB =1. This enables access to registers DLL (Table 15–292) and DLM (Table 15–293) for setting the baud rate. Also, if needed, set the fractional baud rate in the fractional divider register (Table 15–305).
4. UART FIFO: Use bit FIFO enable (bit 0) in register U0FCR (Table 15–297) to enable FIFO.
5. Pins: Select UART pins through PINSEL registers and pin modes through the PINMODE registers (Section 8–5).

   Remark: UART receive pins should not have pull-down resistors enabled.
6. Interrupts: To enable UART interrupts set bit DLAB =0 in register U1LCR (Table 15–298). This enables access to U1IER (Table 15–294). Interrupts are enabled in the NVIC using the appropriate Interrupt Set Enable register.
7. DMA: UART1 transmit and receive functions can operated with the GPDMA controller (see Table 31–544).

2. Features

- Full modem control handshaking available
- Data sizes of 5, 6, 7, and 8 bits.
- Parity generation and checking: odd, even mark, space or none.
- One or two stop bits.
- 16 byte Receive and Transmit FIFOs.
- Built-in baud rate generator, including a fractional rate divider for great versatility.
- Supports DMA for both transmit and receive.
- Auto-baud capability
- Break generation and detection.
- Multiprocessor addressing mode.
- RS-485 support.

3. Pin description

Table 288: UART1 Pin Description

Pin	Type	Description
RXD1	Input	**Serial Input.** Serial receive data.
TXD1	Output	**Serial Output.** Serial transmit data.
CTS1	Input	**Clear To Send.** Active low signal indicates if the external modem is ready to accept transmitted data via TXD1 from the UART1. In normal operation of the modem interface (U1MCR[4] = 0), the complement value of this signal is stored in U1MSR[4]. State change information is stored in U1MSR[0] and is a source for a priority level 4 interrupt, if enabled (U1IER[3] = 1). Clear to send. CTS1 is an asynchronous, active low modem status signal. Its condition can be checked by reading bit 4 (CTS) of the modem status register. Bit 0 (DCTS) of the Modem Status Register (MSR) indicates that CTS1 has changed states since the last read from the MSR. If the modem status interrupt is enabled when CTS1 changes levels and the auto-cts mode is not enabled, an interrupt is generated. CTS1 is also used in the auto-cts mode to control the transmitter.
DCD1	Input	**Data Carrier Detect.** Active low signal indicates if the external modem has established a communication link with the UART1 and data may be exchanged. In normal operation of the modem interface (U1MCR[4]=0), the complement value of this signal is stored in U1MSR[7]. State change information is stored in U1MSR3 and is a source for a priority level 4 interrupt, if enabled (U1IER[3] = 1).
DSR1	Input	**Data Set Ready.** Active low signal indicates if the external modem is ready to establish a communications link with the UART1. In normal operation of the modem interface (U1MCR[4] = 0), the complement value of this signal is stored in U1MSR[5]. State change information is stored in U1MSR[1] and is a source for a priority level 4 interrupt, if enabled (U1IER[3] = 1).
DTR1	Output	Data Terminal Ready. Active low signal indicates that the UART1 is ready to establish connection with external modem. The complement value of this signal is stored in U1MCR[0]. The DTR pin can also be used as an RS-485/EIA-485 output enable signal.
RI1	Input	**Ring Indicator.** Active low signal indicates that a telephone ringing signal has been detected by the modem. In normal operation of the modem interface (U1MCR[4] = 0), the complement value of this signal is stored in U1MSR[6]. State change information is stored in U1MSR[2] and is a source for a priority level 4 interrupt, if enabled (U1IER[3] = 1).
RTS1	Output	**Request To Send.** Active low signal indicates that the UART1 would like to transmit data to the external modem. The complement value of this signal is stored in U1MCR[1]. In auto-rts mode, RTS1 is used to control the transmitter FIFO threshold logic. Request to send. RTS1 is an active low signal informing the modem or data set that the UART is ready to receive data. RTS1 is set to the active (low) level by setting the RTS modem control register bit and is set to the inactive (high) level either as a result of a system reset or during loop-back mode operations or by clearing bit 1 (RTS) of the MCR. In the auto-rts mode, RTS1 is controlled by the transmitter FIFO threshold logic. The RTS pin can also be used as an RS-485/EIA-485 output enable signal.

4. Register description

UART1 contains registers organized as shown in Table 15–289. The Divisor Latch Access Bit (DLAB) is contained in U1LCR[7] and enables access to the Divisor Latches.

Table 289: UART1 register map

Name	Description	Access	Reset Value[1]	Address
U1RBR (when DLAB =0)	Receiver Buffer Register. Contains the next received character to be read.	RO	NA	0x4001 0000 (when DLAB=0)
U1THR (when DLAB =0)	Transmit Holding Register. The next character to be transmitted is written here.	WO	NA	0x4001 0000 (when DLAB=0)
U1DLL (when DLAB =1)	Divisor Latch LSB. Least significant byte of the baud rate divisor value. The full divisor is used to generate a baud rate from the fractional rate divider.	R/W	0x01	0x4001 0000 (when DLAB=1)
U1DLM (when DLAB =1)	Divisor Latch MSB. Most significant byte of the baud rate divisor value. The full divisor is used to generate a baud rate from the fractional rate divider.	R/W	0x00	0x4001 0004 (when DLAB=1)
U1IER (when DLAB =0)	Interrupt Enable Register. Contains individual interrupt enable bits for the 7 potential UART1 interrupts.	R/W	0x00	0x4001 0004 (when DLAB=0)
U1IIR	Interrupt ID Register. Identifies which interrupt(s) are pending.	RO	0x01	0x4001 0008
U1FCR	FIFO Control Register. Controls UART1 FIFO usage and modes.	WO	0x00	0x4001 0008
U1LCR	Line Control Register. Contains controls for frame formatting and break generation.	R/W	0x00	0x4001 000C
U1MCR	Modem Control Register. Contains controls for flow control handshaking and loopback mode.	R/W	0x00	0x4001 0010
U1LSR	Line Status Register. Contains flags for transmit and receive status, including line errors.	RO	0x60	0x4001 0014
U1MSR	Modem Status Register. Contains handshake signal status flags.	RO	0x00	0x4001 0018
U1SCR	Scratch Pad Register. 8-bit temporary storage for software.	R/W	0x00	0x4001 001C
U1ACR	Auto-baud Control Register. Contains controls for the auto-baud feature.	R/W	0x00	0x4001 0020
U1FDR	Fractional Divider Register. Generates a clock input for the baud rate divider.	R/W	0x10	0x4001 0028
U1TER	Transmit Enable Register. Turns off UART transmitter for use with software flow control.	R/W	0x80	0x4001 0030
U1RS485CTRL	RS-485/EIA-485 Control. Contains controls to configure various aspects of RS-485/EIA-485 modes.	R/W	0x00	0x4001 004C
U1ADRMATCH	RS-485/EIA-485 address match. Contains the address match value for RS-485/EIA-485 mode.	R/W	0x00	0x4001 0050
U1RS485DLY	RS-485/EIA-485 direction control delay.	R/W	0x00	0x4001 0054
U1FIFOLVL	FIFO Level register. Provides the current fill levels of the transmit and receive FIFOs.	RO	0x00	0x4001 0058

[1] Reset Value reflects the data stored in used bits only. It does not include reserved bits content.

4.1 UART1 Receiver Buffer Register (U1RBR - 0x4001 0000, when DLAB = 0)

The U1RBR is the top byte of the UART1 RX FIFO. The top byte of the RX FIFO contains the oldest character received and can be read via the bus interface. The LSB (bit 0) represents the “oldest” received data bit. If the character received is less than 8 bits, the unused MSBs are padded with zeroes.

The Divisor Latch Access Bit (DLAB) in U1LCR must be zero in order to access the U1RBR. The U1RBR is always read-only.

Since PE, FE and BI bits correspond to the byte sitting on the top of the RBR FIFO (i.e. the one that will be read in the next read from the RBR), the right approach for fetching the valid pair of received byte and its status bits is first to read the content of the U1LSR register, and then to read a byte from the U1RBR.

Table 290: UART1 Receiver Buffer Register (U1RBR - address 0x4001 0000 when DLAB = 0) bit description

Bit	Symbol	Description	Reset Value
7:0	RBR	The UART1 Receiver Buffer Register contains the oldest received byte in the UART1 RX FIFO.	undefined
31:8	-	Reserved, the value read from a reserved bit is not defined.	NA

4.2 UART1 Transmitter Holding Register (U1THR - 0x4001 0000 when DLAB = 0)

The write-only U1THR is the top byte of the UART1 TX FIFO. The top byte is the newest character in the TX FIFO and can be written via the bus interface. The LSB represents the first bit to transmit.

The Divisor Latch Access Bit (DLAB) in U1LCR must be zero in order to access the U1THR. The U1THR is write-only.

Table 291: UART1 Transmitter Holding Register (U1THR - address 0x4001 0000 when DLAB = 0) bit description

Bit	Symbol	Description	Reset Value
7:0	THR	Writing to the UART1 Transmit Holding Register causes the data to be stored in the UART1 transmit FIFO. The byte will be sent when it reaches the bottom of the FIFO and the transmitter is available.	NA
31:8	-	Reserved, user software should not write ones to reserved bits.	NA

4.3 UART1 Divisor Latch LSB and MSB Registers (U1DLL - 0x4001 0000 and U1DLM - 0x4001 0004, when DLAB = 1)

The UART1 Divisor Latch is part of the UART1 Baud Rate Generator and holds the value used, along with the Fractional Divider, to divide the APB clock (PCLK) in order to produce the baud rate clock, which must be 16x the desired baud rate. The U1DLL and U1DLM registers together form a 16-bit divisor where U1DLL contains the lower 8 bits of the divisor and U1DLM contains the higher 8 bits of the divisor. A 0x0000 value is treated like a 0x0001 value as division by zero is not allowed.The Divisor Latch Access Bit (DLAB) in U1LCR must be one in order to access the UART1 Divisor Latches. Details on how to select the right value for U1DLL and U1DLM can be found later in this chapter, see Section 15–4.16.

Table 292: UART1 Divisor Latch LSB Register (U1DLL - address 0x4001 0000 when DLAB = 1) bit description

Bit	Symbol	Description	Reset Value
7:0	DLLSB	The UART1 Divisor Latch LSB Register, along with the U1DLM register, determines the baud rate of the UART1.	0x01
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 293: UART1 Divisor Latch MSB Register (U1DLM - address 0x4001 0004 when DLAB = 1) bit description

Bit	Symbol	Description	Reset Value
7:0	DLMSB	The UART1 Divisor Latch MSB Register, along with the U1DLL register, determines the baud rate of the UART1.	0x00
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

4.4 UART1 Interrupt Enable Register (U1IER - 0x4001 0004, when DLAB = 0)

The U1IER is used to enable the four UART1 interrupt sources.

Table 294: UART1 Interrupt Enable Register (U1IER - address 0x4001 0004 when DLAB = 0) bit description

Bit	Symbol	Value	Description	Reset Value
0	RBR Interrupt Enable		enables the Receive Data Available interrupt for UART1. It also controls the Character Receive Time-out interrupt.	0
		0	Disable the RDA interrupts.	
		1	Enable the RDA interrupts.	
1	THRE Interrupt Enable		enables the THRE interrupt for UART1. The status of this interrupt can be read from U1LSR[5].	0
		0	Disable the THRE interrupts.	
		1	Enable the THRE interrupts.	
2	RX Line Interrupt Enable		enables the UART1 RX line status interrupts. The status of this interrupt can be read from U1LSR[4:1].	0
		0	Disable the RX line status interrupts.	
		1	Enable the RX line status interrupts.	
3	Modem Status Interrupt Enable		enables the modem interrupt. The status of this interrupt can be read from U1MSR[3:0].	0
		0	Disable the modem interrupt.	
		1	Enable the modem interrupt.	
6:4	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7	CTS Interrupt Enable		If auto-cts mode is enabled this bit enables/disables the modem status interrupt generation on a CTS1 signal transition. If auto-cts mode is disabled a CTS1 transition will generate an interrupt if Modem Status Interrupt Enable (U1IER[3]) is set. In normal operation a CTS1 signal transition will generate a Modem Status Interrupt unless the interrupt has been disabled by clearing the U1IER[3] bit in the U1IER register. In auto-cts mode a transition on the CTS1 bit will trigger an interrupt only if both the U1IER[3] and U1IER[7] bits are set.	0
		0	Disable the CTS interrupt.	
		1	Enable the CTS interrupt.	

Table 294: UART1 Interrupt Enable Register (U1IER - address 0x4001 0004 when DLAB = 0) bit description

Bit	Symbol	Value	Description	Reset Value
8	ABEOIntEn		Enables the end of auto-baud interrupt.	0
		0	Disable end of auto-baud Interrupt.	
		1	Enable end of auto-baud Interrupt.	
9	ABTOIntEn		Enables the auto-baud time-out interrupt.	0
		0	Disable auto-baud time-out Interrupt.	
		1	Enable auto-baud time-out Interrupt.	
31:10	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

4.5 UART1 Interrupt Identification Register (U1IIR - 0x4001 0008)

The U1IIR provides a status code that denotes the priority and source of a pending interrupt. The interrupts are frozen during an U1IIR access. If an interrupt occurs during an U1IIR access, the interrupt is recorded for the next U1IIR access.

Table 295: UART1 Interrupt Identification Register (U1IIR - address 0x4001 0008) bit description

Bit	Symbol	Value	Description	Reset Value
0	IntStatus		Interrupt status. Note that U1IIR[0] is active low. The pending interrupt can be determined by evaluating U1IIR[3:1].	1
		0	At least one interrupt is pending.	
		1	No interrupt is pending.	
3:1	IntId		Interrupt identification. U1IER[3:1] identifies an interrupt corresponding to the UART1 Rx or TX FIFO. All other combinations of U1IER[3:1] not listed below are reserved (100,101,111).	0
		011	1 - Receive Line Status (RLS).	
		010	2a - Receive Data Available (RDA).	
		110	2b - Character Time-out Indicator (CTI).	
		001	3 - THRE Interrupt.	
		000	4 - Modem Interrupt.	
5:4	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7:6	FIFO Enable		Copies of U1FCR[0].	0
8	ABEOInt		End of auto-baud interrupt. True if auto-baud has finished successfully and interrupt is enabled.	0
9	ABTOInt		Auto-baud time-out interrupt. True if auto-baud has timed out and interrupt is enabled.	0
31:10	-		Reserved, the value read from a reserved bit is not defined.	NA

Bit U1IIR[9:8] are set by the auto-baud function and signal a time-out or end of auto-baud condition. The auto-baud interrupt conditions are cleared by setting the corresponding Clear bits in the Auto-baud Control Register.

If the IntStatus bit is 1 no interrupt is pending and the IntId bits will be zero. If the IntStatus is 0, a non auto-baud interrupt is pending in which case the IntId bits identify the type of interrupt and handling as described in Table 15–296. Given the status of U1IIR[3:0], an interrupt handler routine can determine the cause of the interrupt and how to clear the active interrupt. The U1IIR must be read in order to clear the interrupt prior to exiting the Interrupt Service Routine.

The UART1 RLS interrupt (U1IIR[3:1] = 011) is the highest priority interrupt and is set whenever any one of four error conditions occur on the UART1RX input: overrun error (OE), parity error (PE), framing error (FE) and break interrupt (BI). The UART1 Rx error condition that set the interrupt can be observed via U1LSR[4:1]. The interrupt is cleared upon an U1LSR read.

The UART1 RDA interrupt (U1IIR[3:1] = 010) shares the second level priority with the CTI interrupt (U1IIR[3:1] = 110). The RDA is activated when the UART1 Rx FIFO reaches the trigger level defined in U1FCR7:6 and is reset when the UART1 Rx FIFO depth falls below the trigger level. When the RDA interrupt goes active, the CPU can read a block of data defined by the trigger level.

The CTI interrupt (U1IIR[3:1] = 110) is a second level interrupt and is set when the UART1 Rx FIFO contains at least one character and no UART1 Rx FIFO activity has occurred in 3.5 to 4.5 character times. Any UART1 Rx FIFO activity (read or write of UART1 RSR) will clear the interrupt. This interrupt is intended to flush the UART1 RBR after a message has been received that is not a multiple of the trigger level size. For example, if a peripheral wished to send a 105 character message and the trigger level was 10 characters, the CPU would receive 10 RDA interrupts resulting in the transfer of 100 characters and 1 to 5 CTI interrupts (depending on the service routine) resulting in the transfer of the remaining 5 characters.

Table 296: UART1 Interrupt Handling

U1IIR[3:0] value[1]	Priority	Interrupt Type	Interrupt Source	Interrupt Reset
0001	-	None	None	-
0110	Highest	RX Line Status / Error	OE[2] or PE[2] or FE[2] or BI[2]	U1LSR Read[2]
0100	Second	RX Data Available	Rx data available or trigger level reached in FIFO (U1FCR0=1)	U1RBR Read[3] or UART1 FIFO drops below trigger level
1100	Second	Character Time-out indication	Minimum of one character in the RX FIFO and no character input or removed during a time period depending on how many characters are in FIFO and what the trigger level is set at (3.5 to 4.5 character times). The exact time will be: [(word length) $\times$ 7 - 2] $\times$ 8 + [(trigger level - number of characters) $\times$ 8 + 1] RCLKs	U1RBR Read[3]
0010	Third	THRE	THRE[2]	U1IIR Read[4] (if source of interrupt) or THR write
0000	Fourth	Modem Status	CTS or DSR or RI or DCD	MSR Read

[1] Values "0000", “0011”, “0101”, “0111”, “1000”, “1001”, “1010”, “1011”,”1101”,”1110”,”1111” are reserved.

[2] For details see Section 15–4.10 “UART1 Line Status Register (U1LSR - 0x4001 0014)”

[3] For details see Section 15–4.1 “UART1 Receiver Buffer Register (U1RBR - 0x4001 0000, when DLAB = 0)”

[4] For details see Section 15–4.5 “UART1 Interrupt Identification Register (U1IIR - 0x4001 0008)” and Section 15–4.2 “UART1 Transmitter Holding Register (U1THR - 0x4001 0000 when DLAB = 0)”

The UART1 THRE interrupt (U1IIR[3:1] = 001) is a third level interrupt and is activated when the UART1 THR FIFO is empty provided certain initialization conditions have been met. These initialization conditions are intended to give the UART1 THR FIFO a chance to fill up with data to eliminate many THRE interrupts from occurring at system start-up. The initialization conditions implement a one character delay minus the stop bit whenever THRE = 1 and there have not been at least two characters in the U1THR at one time since the last THRE = 1 event. This delay is provided to give the CPU time to write data to U1THR without a THRE interrupt to decode and service. A THRE interrupt is set immediately if the UART1 THR FIFO has held two or more characters at one time and currently, the U1THR is empty. The THRE interrupt is reset when a U1THR write occurs or a read of the U1IIR occurs and the THRE is the highest interrupt (U1IIR[3:1] = 001).

It is the lowest priority interrupt and is activated whenever there is any state change on modem inputs pins, DCD, DSR or CTS. In addition, a low to high transition on modem input RI will generate a modem interrupt. The source of the modem interrupt can be determined by examining U1MSR[3:0]. A U1MSR read will clear the modem interrupt.

4.6 UART1 FIFO Control Register (U1FCR - 0x4001 0008)

The write-only U1FCR controls the operation of the UART1 RX and TX FIFOs.

Table 297: UART1 FIFO Control Register (U1FCR - address 0x4001 0008) bit description

Bit	Symbol	Value	Description	Reset Value
0	FIFO Enable	0	UART1 FIFOs are disabled. Must not be used in the application.	0
		1	Active high enable for both UART1 Rx and TX FIFOs and U1FCR[7:1] access. This bit must be set for proper UART1 operation. Any transition on this bit will automatically clear the UART1 FIFOs.	
1	RX FIFO Reset	0	No impact on either of UART1 FIFOs.	0
		1	Writing a logic 1 to U1FCR[1] will clear all bytes in UART1 Rx FIFO, reset the pointer logic. This bit is self-clearing.	
2	TX FIFO Reset	0	No impact on either of UART1 FIFOs.	0
		1	Writing a logic 1 to U1FCR[2] will clear all bytes in UART1 TX FIFO, reset the pointer logic. This bit is self-clearing.	
3	DMA Mode Select		When the FIFO enable bit (bit 0 of this register) is set, this bit selects the DMA mode. See Section 15–4.6.1.	0
5:4	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7:6	RX Trigger Level		These two bits determine how many receiver UART1 FIFO characters must be written before an interrupt is activated.	0
		00	Trigger level 0 (1 character or 0x01).	
		01	Trigger level 1 (4 characters or 0x04).	
		10	Trigger level 2 (8 characters or 0x08).	
		11	Trigger level 3 (14 characters or 0x0E).	
31:8	-		Reserved, user software should not write ones to reserved bits.	NA

4.6.1 DMA Operation

The user can optionally operate the UART transmit and/or receive using DMA. The DMA mode is determined by the DMA Mode Select bit in the FCR register. This bit only has an affect when the FIFOs are enabled via the FIFO Enable bit in the FCR register.

UART receiver DMA

In DMA mode, the receiver DMA request is asserted on the event of the receiver FIFO level becoming equal to or greater than trigger level, or if a character timeout occurs. See the description of the RX Trigger Level above. The receiver DMA request is cleared by the DMA controller.

UART transmitter DMA

In DMA mode, the transmitter DMA request is asserted on the event of the transmitter FIFO transitioning to not full. The transmitter DMA request is cleared by the DMA controller.

4.7 UART1 Line Control Register (U1LCR - 0x4001 000C)

The U1LCR determines the format of the data character that is to be transmitted or received.

Table 298: UART1 Line Control Register (U1LCR - address 0x4001 000C) bit description

Bit	Symbol	Value	Description	Reset Value
1:0	Word Length Select	00	5-bit character length.	0
		01	6-bit character length.	
		10	7-bit character length.	
		11	8-bit character length.	
2	Stop Bit Select	0	1 stop bit.	0
		1	2 stop bits (1.5 if U1LCR[1:0]=00).	
3	Parity Enable	0	Disable parity generation and checking.	0
		1	Enable parity generation and checking.	
5:4	Parity Select	00	Odd parity. Number of 1s in the transmitted character and the attached parity bit will be odd.	0
		01	Even Parity. Number of 1s in the transmitted character and the attached parity bit will be even.	
		10	Forced "1" stick parity.	
		11	Forced "0" stick parity.	
6	Break Control	0	Disable break transmission.	0
		1	Enable break transmission. Output pin UART1 TXD is forced to logic 0 when U1LCR[6] is active high.	
7	Divisor Latch Access Bit (DLAB)	0	Disable access to Divisor Latches.	0
		1	Enable access to Divisor Latches.	
31:8	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

4.8 UART1 Modem Control Register (U1MCR - 0x4001 0010)

The U1MCR enables the modem loopback mode and controls the modem output signals.

Table 299: UART1 Modem Control Register (U1MCR - address 0x4001 0010) bit description

Bit	Symbol	Value	Description	Reset value
0	DTR Control		Source for modem output pin, DTR. This bit reads as 0 when modem loopback mode is active.	0
1	RTS Control		Source for modem output pin RTS. This bit reads as 0 when modem loopback mode is active.	0
3-2	-	NA	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	0
4	Loopback Mode Select		The modem loopback mode provides a mechanism to perform diagnostic loopback testing. Serial data from the transmitter is connected internally to serial input of the receiver. Input pin, RXD1, has no effect on loopback and output pin, TXD1 is held in marking state. The 4 modem inputs (CTS, DSR, RI and DCD) are disconnected externally. Externally, the modem outputs (RTS, DTR) are set inactive. Internally, the 4 modem outputs are connected to the 4 modem inputs. As a result of these connections, the upper 4 bits of the U1MSR will be driven by the lower 4 bits of the U1MCR rather than the 4 modem inputs in normal mode. This permits modem status interrupts to be generated in loopback mode by writing the lower 4 bits of U1MCR.	0
		0	Disable modem loopback mode.	
		1	Enable modem loopback mode.	
5	-	NA	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	0
6	RTSen	0	Disable auto-rts flow control.	0
		1	Enable auto-rts flow control.	
7	CTSen	0	Disable auto-cts flow control.	0
		1	Enable auto-cts flow control.	
31:8	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

4.9 Auto-flow control

If auto-RTS mode is enabled the UART1‘s receiver FIFO hardware controls the RTS1 output of the UART1. If the auto-CTS mode is enabled the UART1‘s U1TSR hardware will only start transmitting if the CTS1 input signal is asserted.

15.4.9.1 Auto-RTS

The auto-RTS function is enabled by setting the RTSen bit. Auto-RTS data flow control originates in the U1RBR module and is linked to the programmed receiver FIFO trigger level. If auto-RTS is enabled, the data-flow is controlled as follows:

When the receiver FIFO level reaches the programmed trigger level, RTS1 is de-asserted (to a high value). It is possible that the sending UART sends an additional byte after the trigger level is reached (assuming the sending UART has another byte to send) because it might not recognize the de-assertion of RTS1 until after it has begun sending the additional byte. RTS1 is automatically reasserted (to a low value) once the receiver FIFO has reached the previous trigger level. The re-assertion of RTS1 signals to the sending UART to continue transmitting data.

If Auto-RTS mode is disabled, the RTSen bit controls the RTS1 output of the UART1. If Auto-RTS mode is enabled, hardware controls the RTS1 output, and the actual value of RTS1 will be copied in the RTS Control bit of the UART1. As long as Auto-RTS is enabled, the value of the RTS Control bit is read-only for software.

Example: Suppose the UART1 operating in ‘550 mode has trigger level in U1FCR set to 0x2 then if Auto-RTS is enabled the UART1 will de-assert the RTS1 output as soon as the receive FIFO contains 8 bytes (Table 15–297 on page 323). The RTS1 output will be reasserted as soon as the receive FIFO hits the previous trigger level: 4 bytes.

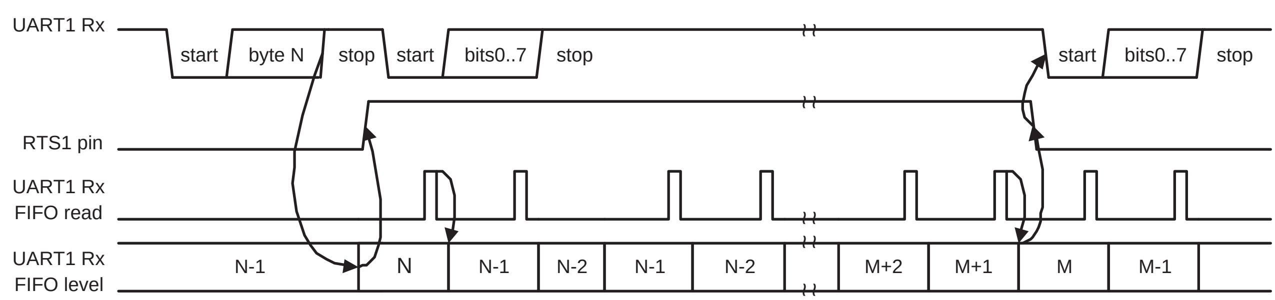


Fig 47. Auto-RTS Functional Timing

15.4.9.2 Auto-CTS

The Auto-CTS function is enabled by setting the CTSen bit. If Auto-CTS is enabled the transmitter circuitry in the U1TSR module checks CTS1 input before sending the next data byte. When CTS1 is active (low), the transmitter sends the next byte. To stop the transmitter from sending the following byte, CTS1 must be released before the middle of the last stop bit that is currently being sent. In Auto-CTS mode a change of the CTS1 signal does not trigger a modem status interrupt unless the CTS Interrupt Enable bit is set, Delta CTS bit in the U1MSR will be set though. Table 15–300 lists the conditions for generating a Modem Status interrupt.

Table 300: Modem status interrupt generation

Enable Modem Status Interrupt (U1ER[3])	CTSen (U1MCR[7])	CTS Interrupt Enable (U1IER[7])	Delta CTS (U1MSR[0])	Delta DCD or Trailing Edge RI or Delta DSR (U1MSR[3] or U1MSR[2] or U1MSR[1])	Modem Status Interrupt
0	x	x	x	x	No
1	0	x	0	0	No
1	0	x	1	x	Yes
1	0	x	x	1	Yes
1	1	0	x	0	No
1	1	0	x	1	Yes
1	1	1	0	0	No
1	1	1	1	x	Yes
1	1	1	x	1	Yes

The auto-CTS function reduces interrupts to the host system. When flow control is enabled, a CTS1 state change does not trigger host interrupts because the device automatically controls its own transmitter. Without Auto-CTS, the transmitter sends any data present in the transmit FIFO and a receiver overrun error can result. Figure 15–48 illustrates the Auto-CTS functional timing.

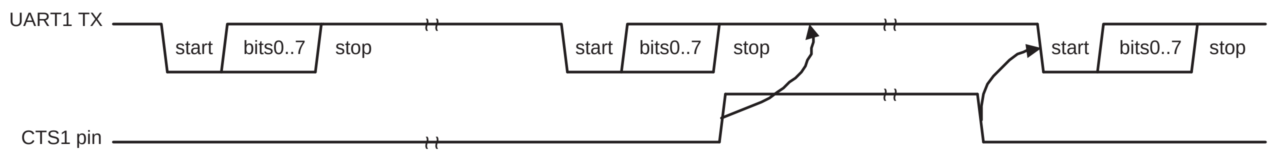


Fig 48. Auto-CTS Functional Timing

While starting transmission of the initial character the CTS1 signal is asserted. Transmission will stall as soon as the pending transmission has completed. The UART will continue transmitting a 1 bit as long as CTS1 is de-asserted (high). As soon as CTS1 gets de-asserted transmission resumes and a start bit is sent followed by the data bits of the next character.

4.10 UART1 Line Status Register (U1LSR - 0x4001 0014)

The U1LSR is a read-only register that provides status information on the UART1 TX and RX blocks.

Table 301: UART1 Line Status Register (U1LSR - address 0x4001 0014) bit description

Bit	Symbol	Value	Description	Reset Value
0	Receiver Data Ready (RDR)		U1LSR[0] is set when the U1RBR holds an unread character and is cleared when the UART1 RBR FIFO is empty.	0
		0	The UART1 receiver FIFO is empty.	
		1	The UART1 receiver FIFO is not empty.	
1	Overrun Error (OE)		The overrun error condition is set as soon as it occurs. An U1LSR read clears U1LSR[1]. U1LSR[1] is set when UART1 RSR has a new character assembled and the UART1 RBR FIFO is full. In this case, the UART1 RBR FIFO will not be overwritten and the character in the UART1 RSR will be lost.	0
		0	Overrun error status is inactive.	
		1	Overrun error status is active.	
2	Parity Error (PE)		When the parity bit of a received character is in the wrong state, a parity error occurs. An U1LSR read clears U1LSR[2]. Time of parity error detection is dependent on U1FCR[0]. **Note:** A parity error is associated with the character at the top of the UART1 RBR FIFO.	0
		0	Parity error status is inactive.	
		1	Parity error status is active.	

Table 301: UART1 Line Status Register (U1LSR - address 0x4001 0014) bit description

Bit	Symbol	Value	Description	Reset Value
3	Framing Error (FE)		When the stop bit of a received character is a logic 0, a framing error occurs. An U1LSR read clears U1LSR[3]. The time of the framing error detection is dependent on U1FCR0. Upon detection of a framing error, the RX will attempt to resynchronize to the data and assume that the bad stop bit is actually an early start bit. However, it cannot be assumed that the next received byte will be correct even if there is no Framing Error. **Note:** A framing error is associated with the character at the top of the UART1 RBR FIFO.	0
		0	Framing error status is inactive.	
		1	Framing error status is active.	
4	Break Interrupt (BI)		When RXD1 is held in the spacing state (all zeroes) for one full character transmission (start, data, parity, stop), a break interrupt occurs. Once the break condition has been detected, the receiver goes idle until RXD1 goes to marking state (all ones). An U1LSR read clears this status bit. The time of break detection is dependent on U1FCR[0]. **Note:** The break interrupt is associated with the character at the top of the UART1 RBR FIFO.	0
		0	Break interrupt status is inactive.	
		1	Break interrupt status is active.	
5	Transmitter Holding Register Empty (THRE)		THRE is set immediately upon detection of an empty UART1 THR and is cleared on a U1THR write.	1
		0	U1THR contains valid data.	
		1	U1THR is empty.	
6	Transmitter Empty (TEMT)		TEMT is set when both U1THR and U1TSR are empty; TEMT is cleared when either the U1TSR or the U1THR contain valid data.	1
		0	U1THR and/or the U1TSR contains valid data.	
		1	U1THR and the U1TSR are empty.	
7	Error in RX FIFO (RXFE)		U1LSR[7] is set when a character with a RX error such as framing error, parity error or break interrupt, is loaded into the U1RBR. This bit is cleared when the U1LSR register is read and there are no subsequent errors in the UART1 FIFO.	0
		0	U1RBR contains no UART1 RX errors or U1FCR[0]=0.	
		1	UART1 RBR contains at least one UART1 RX error.	
31:8	-		Reserved, the value read from a reserved bit is not defined.	NA

4.11 UART1 Modem Status Register (U1MSR - 0x4001 0018)

The U1MSR is a read-only register that provides status information on the modem input signals. U1MSR[3:0] is cleared on U1MSR read. Note that modem signals have no direct effect on UART1 operation, they facilitate software implementation of modem signal operations.

Table 302: UART1 Modem Status Register (U1MSR - address 0x4001 0018) bit description

Bit	Symbol	Value	Description	Reset Value
0	Delta CTS		Set upon state change of input CTS. Cleared on an U1MSR read.	0
		0	No change detected on modem input, CTS.	
		1	State change detected on modem input, CTS.	

Table 302: UART1 Modem Status Register (U1MSR - address 0x4001 0018) bit description

Bit	Symbol	Value	Description	Reset Value
1	Delta DSR		Set upon state change of input DSR. Cleared on an U1MSR read.	0
		0	No change detected on modem input, DSR.	
		1	State change detected on modem input, DSR.	
2	Trailing Edge RI		Set upon low to high transition of input RI. Cleared on an U1MSR read.	0
		0	No change detected on modem input, RI.	
		1	Low-to-high transition detected on RI.	
3	Delta DCD		Set upon state change of input DCD. Cleared on an U1MSR read.	0
		0	No change detected on modem input, DCD.	
		1	State change detected on modem input, DCD.	
4	CTS		Clear To Send State. Complement of input signal CTS. This bit is connected to U1MCR[1] in modem loopback mode.	0
5	DSR		Data Set Ready State. Complement of input signal DSR. This bit is connected to U1MCR[0] in modem loopback mode.	0
6	RI		Ring Indicator State. Complement of input RI. This bit is connected to U1MCR[2] in modem loopback mode.	0
7	DCD		Data Carrier Detect State. Complement of input DCD. This bit is connected to U1MCR[3] in modem loopback mode.	0
31:8	-		Reserved, the value read from a reserved bit is not defined.	NA

4.12 UART1 Scratch Pad Register (U1SCR - 0x4001 001C)

The U1SCR has no effect on the UART1 operation. This register can be written and/or read at user's discretion. There is no provision in the interrupt interface that would indicate to the host that a read or write of the U1SCR has occurred.

Table 303: UART1 Scratch Pad Register (U1SCR - address 0x4001 0014) bit description

Bit	Symbol	Description	Reset Value
7:0	Pad	A readable, writable byte.	0x00

4.13 UART1 Auto-baud Control Register (U1ACR - 0x4001 0020)

The UART1 Auto-baud Control Register (U1ACR) controls the process of measuring the incoming clock/data rate for the baud rate generation and can be read and written at user's discretion.

Table 304: Auto-baud Control Register (U1ACR - address 0x4001 0020) bit description

Bit	Symbol	Value	Description	Reset value
0	Start		This bit is automatically cleared after auto-baud completion.	0
		0	Auto-baud stop (auto-baud is not running).	
		1	Auto-baud start (auto-baud is running). Auto-baud run bit. This bit is automatically cleared after auto-baud completion.	
1	Mode		Auto-baud mode select bit.	0
		0	Mode 0.	
		1	Mode 1.	

Table 304: Auto-baud Control Register (U1ACR - address 0x4001 0020) bit description

Bit	Symbol	Value	Description	Reset value
2	AutoRestart	0	No restart	0
		1	Restart in case of time-out (counter restarts at next UART1 Rx falling edge)	0
7:3	-	NA	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	0
8	ABEOIntClr		End of auto-baud interrupt clear bit (write-only accessible).	0
		0	Writing a 0 has no impact.	
		1	Writing a 1 will clear the corresponding interrupt in the U1IIR.	
9	ABTOIntClr		Auto-baud time-out interrupt clear bit (write-only accessible).	0
		0	Writing a 0 has no impact.	
		1	Writing a 1 will clear the corresponding interrupt in the U1IIR.	
31:10	-	NA	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	0

4.14 Auto-baud

The UART1 auto-baud function can be used to measure the incoming baud-rate based on the “AT” protocol (Hayes command). If enabled the auto-baud feature will measure the bit time of the receive data stream and set the divisor latch registers U1DLM and U1DLL accordingly.

Remark: the fractional rate divider is not connected during auto-baud operations, and therefore should not be used when the auto-baud feature is needed.

Auto-baud is started by setting the U1ACR Start bit. Auto-baud can be stopped by clearing the U1ACR Start bit. The Start bit will clear once auto-baud has finished and reading the bit will return the status of auto-baud (pending/finished).

Two auto-baud measuring modes are available which can be selected by the U1ACR Mode bit. In mode 0 the baud-rate is measured on two subsequent falling edges of the UART1 Rx pin (the falling edge of the start bit and the falling edge of the least significant bit). In mode 1 the baud-rate is measured between the falling edge and the subsequent rising edge of the UART1 Rx pin (the length of the start bit).

The U1ACR AutoRestart bit can be used to automatically restart baud-rate measurement if a time-out occurs (the rate measurement counter overflows). If this bit is set the rate measurement will restart at the next falling edge of the UART1 Rx pin.

The auto-baud function can generate two interrupts.

- The U1IIR ABTOInt interrupt will get set if the interrupt is enabled (U1IER ABToIntEn is set and the auto-baud rate measurement counter overflows).
- The U1IIR ABEOInt interrupt will get set if the interrupt is enabled (U1IER ABEOIntEn is set and the auto-baud has completed successfully).

The auto-baud interrupts have to be cleared by setting the corresponding U1ACR ABTOIntClr and ABEOIntEn bits.

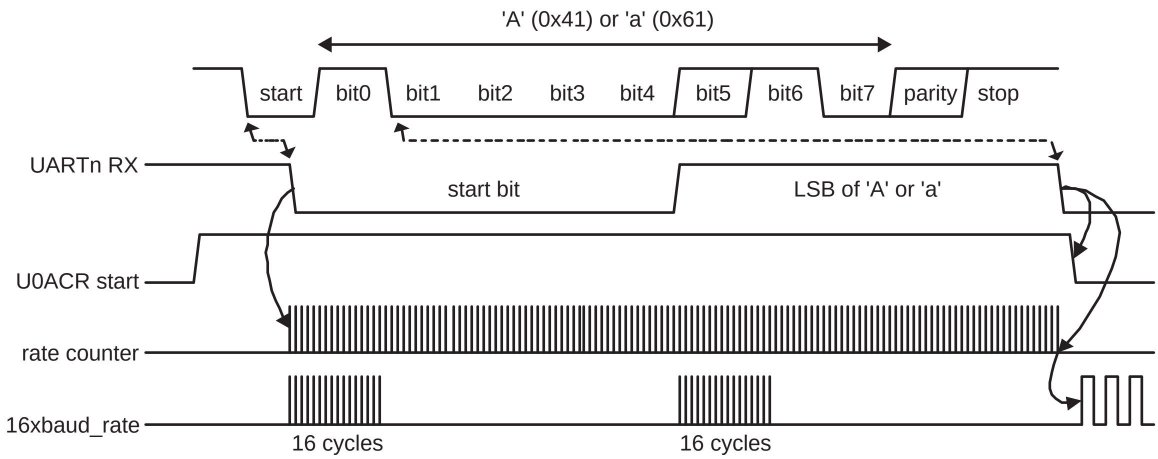
Typically the fractional baud-rate generator is disabled (DIVADDVAL = 0) during auto-baud. However, if the fractional baud-rate generator is enabled (DIVADDVAL > 0), it is going to impact the measuring of UART1 Rx pin baud-rate, but the value of the U1FDR register is not going to be modified after rate measurement. Also, when auto-baud is used, any write to U1DLM and U1DLL registers should be done before U1ACR register write. The minimum and the maximum baud rates supported by UART1 are function of pclk, number of data bits, stop bits and parity bits.

$$ratemin = \frac{2 \times PCLK}{16 \times 2^{15}} \leq UART1_{baudrate} \leq \frac{PCLK}{16 \times (2 + databits + paritybits + stopbits)} = ratemax \quad (3)$$

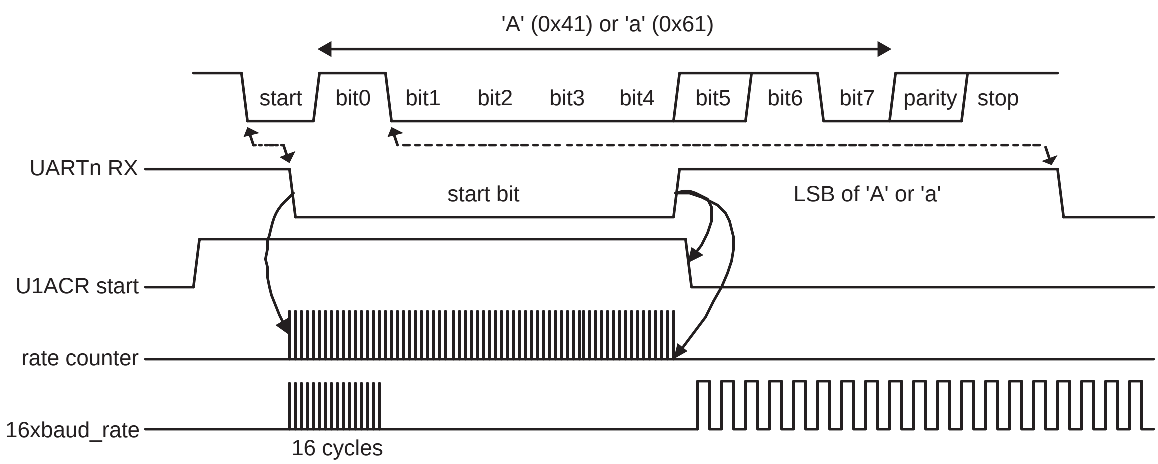
4.15 Auto-baud modes

When the software is expecting an “AT” command, it configures the UART1 with the expected character format and sets the U1ACR Start bit. The initial values in the divisor latches U1DLM and U1DLM don‘t care. Because of the “A” or “a” ASCII coding (”A" = 0x41, “a” = 0x61), the UART1 Rx pin sensed start bit and the LSB of the expected character are delimited by two falling edges. When the U1ACR Start bit is set, the auto-baud protocol will execute the following phases:

1. On U1ACR Start bit setting, the baud-rate measurement counter is reset and the UART1 U1RSR is reset. The U1RSR baud rate is switch to the highest rate.
2. A falling edge on UART1 Rx pin triggers the beginning of the start bit. The rate measuring counter will start counting pclk cycles optionally pre-scaled by the fractional baud-rate generator.
3. During the receipt of the start bit, 16 pulses are generated on the RSR baud input with the frequency of the (fractional baud-rate pre-scaled) UART1 input clock, guaranteeing the start bit is stored in the U1RSR.
4. During the receipt of the start bit (and the character LSB for mode = 0) the rate counter will continue incrementing with the pre-scaled UART1 input clock (pclk).
5. If Mode = 0 then the rate counter will stop on next falling edge of the UART1 Rx pin. If Mode = 1 then the rate counter will stop on the next rising edge of the UART1 Rx pin.
6. The rate counter is loaded into U1DLM/U1DLL and the baud-rate will be switched to normal operation. After setting the U1DLM/U1DLL the end of auto-baud interrupt U1IIR ABEOInt will be set, if enabled. The U1RSR will now continue receiving the remaining bits of the “A/a” character.



a. Mode 0 (start bit and LSB are used for auto-baud)



b. Mode 1 (only start bit is used for auto-baud)

Fig 49. Auto-baud a) mode 0 and b) mode 1 waveform

4.16 UART1 Fractional Divider Register (U1FDR - 0x4001 0028)

The UART1 Fractional Divider Register (U1FDR) controls the clock pre-scaler for the baud rate generation and can be read and written at the user’s discretion. This pre-scaler takes the APB clock and generates an output clock according to the specified fractional requirements.

Important: If the fractional divider is active (DIVADDVAL > 0) and DLM = 0, the value of the DLL register must be greater than 2.

Table 305: UART1 Fractional Divider Register (U1FDR - address 0x4001 0028) bit description

Bit	Function	Value	Description	Reset value
3:0	DIVADDVAL	0	Baud-rate generation pre-scaler divisor value. If this field is 0, fractional baud-rate generator will not impact the UARTn baudrate.	0
7:4	MULVAL	1	Baud-rate pre-scaler multiplier value. This field must be greater or equal 1 for UARTn to operate properly, regardless of whether the fractional baud-rate generator is used or not.	1
31:8	-	NA	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	0

This register controls the clock pre-scaler for the baud rate generation. The reset value of the register keeps the fractional capabilities of UART1 disabled making sure that UART1 is fully software and hardware compatible with UARTs not equipped with this feature.

UART1 baud rate can be calculated as (n = 1):

$$UART1_{baudrate} = \frac{PCLK}{16 \times (256 \times U1DLM + U1DLL) \times \left(1 + \frac{DivAddVal}{MulVal}\right)} \tag{4}$$

Where PCLK is the peripheral clock, U1DLM and U1DLL are the standard UART1 baud rate divider registers, and DIVADDVAL and MULVAL are UART1 fractional baud rate generator specific parameters.

The value of MULVAL and DIVADDVAL should comply to the following conditions:

1. 1 ≤ MULVAL ≤ 15
2. 0 ≤ DIVADDVAL ≤ 14
3. DIVADDVAL < MULVAL

The value of the U1FDR should not be modified while transmitting/receiving data or data may be lost or corrupted.

If the U1FDR register value does not comply to these two requests, then the fractional divider output is undefined. If DIVADDVAL is zero then the fractional divider is disabled, and the clock will not be divided.

4.16.1 Baud rate calculation

UART1 can operate with or without using the Fractional Divider. In real-life applications it is likely that the desired baud rate can be achieved using several different Fractional Divider settings. The following algorithm illustrates one way of finding a set of DLM, DLL, MULVAL, and DIVADDVAL values. Such set of parameters yields a baud rate with a relative error of less than 1.1% from the desired one.

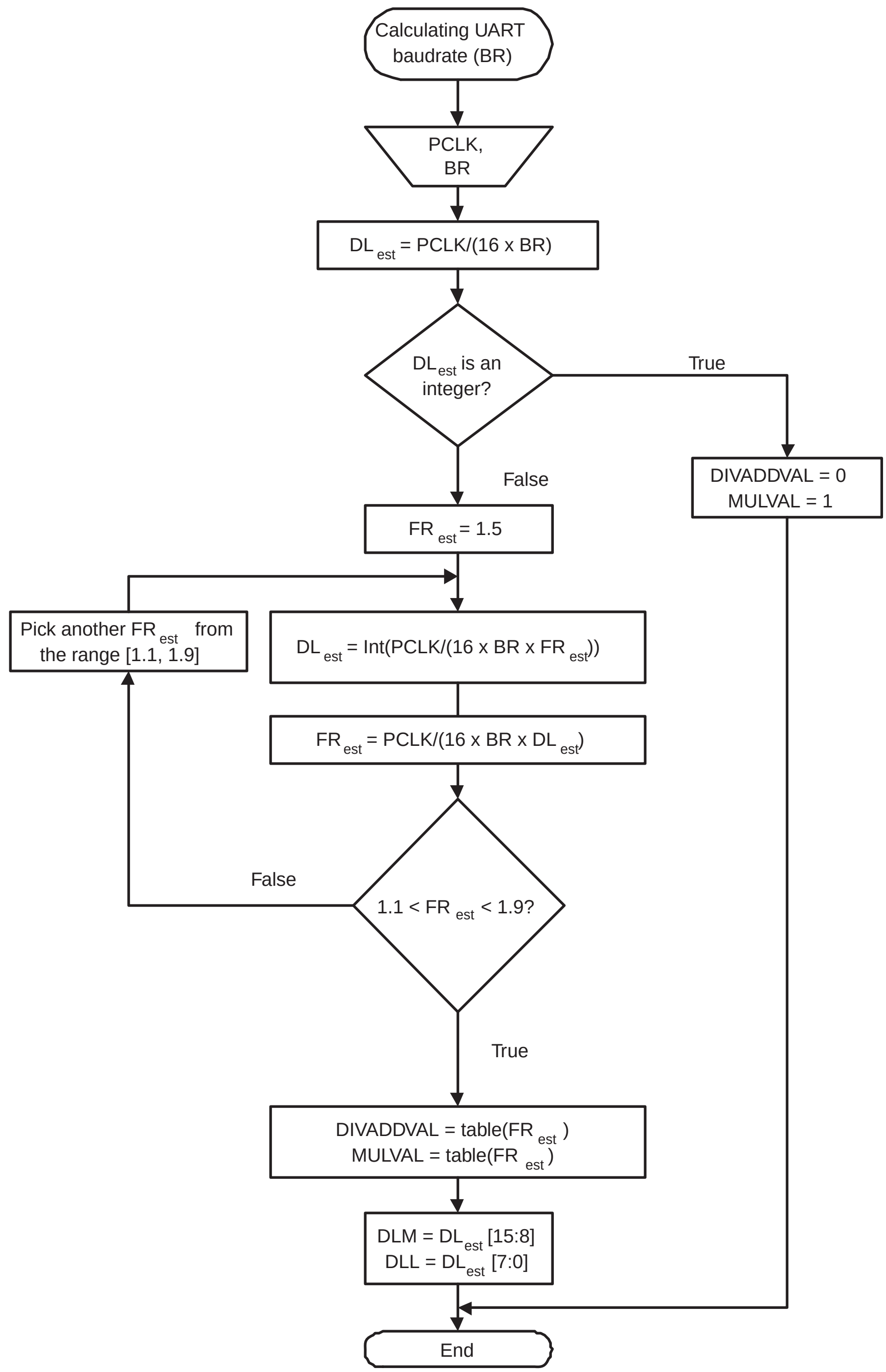


Fig 50. Algorithm for setting UART dividers

Table 306. Fractional Divider setting look-up table

FR	DivAddVal/ MulVal	FR	DivAddVal/ MulVal	FR	DivAddVal/ MulVal	FR	DivAddVal/ MulVal
1.000	0/1	1.250	1/4	1.500	1/2	1.750	3/4
1.067	1/15	1.267	4/15	1.533	8/15	1.769	10/13
1.071	1/14	1.273	3/11	1.538	7/13	1.778	7/9
1.077	1/13	1.286	2/7	1.545	6/11	1.786	11/14
1.083	1/12	1.300	3/10	1.556	5/9	1.800	4/5
1.091	1/11	1.308	4/13	1.571	4/7	1.818	9/11
1.100	1/10	1.333	1/3	1.583	7/12	1.833	5/6
1.111	1/9	1.357	5/14	1.600	3/5	1.846	11/13
1.125	1/8	1.364	4/11	1.615	8/13	1.857	6/7
1.133	2/15	1.375	3/8	1.625	5/8	1.867	13/15
1.143	1/7	1.385	5/13	1.636	7/11	1.875	7/8
1.154	2/13	1.400	2/5	1.643	9/14	1.889	8/9
1.167	1/6	1.417	5/12	1.667	2/3	1.900	9/10
1.182	2/11	1.429	3/7	1.692	9/13	1.909	10/11
1.200	1/5	1.444	4/9	1.700	7/10	1.917	11/12
1.214	3/14	1.455	5/11	1.714	5/7	1.923	12/13
1.222	2/9	1.462	6/13	1.727	8/11	1.929	13/14
1.231	3/13	1.467	7/15	1.733	11/15	1.933	14/15

4.16.1.1 Example 1: PCLK = 14.7456 MHz, BR = 9600

According to the provided algorithm DL_{est} = PCLK/(16 x BR) = 14.7456 MHz / (16 x 9600) = 96. Since this DL_{est} is an integer number, DIVADDVAL = 0, MULVAL = 1, DLM = 0, and DLL = 96.

4.16.1.2 Example 2: PCLK = 12 MHz, BR = 115200

According to the provided algorithm DL_{est} = PCLK/(16 x BR) = 12 MHz / (16 x 115200) = 6.51. This DL_{est} is not an integer number and the next step is to estimate the FR parameter. Using an initial estimate of FR_{est} = 1.5 a new DL_{est} = 4 is calculated and FR_{est} is recalculated as FR_{est} = 1.628. Since FRest = 1.628 is within the specified range of 1.1 and 1.9, DIVADDVAL and MULVAL values can be obtained from the attached look-up table.

The closest value for FRest = 1.628 in the look-up Table 15–306 is FR = 1.625. It is equivalent to DIVADDVAL = 5 and MULVAL = 8.

Based on these findings, the suggested UART setup would be: DLM = 0, DLL = 4, DIVADDVAL = 5, and MULVAL = 8. According to Equation 15–4 the UART rate is 115384. This rate has a relative error of 0.16% from the originally specified 115200.

4.17 UART1 Transmit Enable Register (U1TER - 0x4001 0030)

In addition to being equipped with full hardware flow control (auto-cts and auto-rts mechanisms described above), U1TER enables implementation of software flow control, too. When TxEn=1, UART1 transmitter will keep sending data as long as they are available. As soon as TxEn becomes 0, UART1 transmission will stop.

Although Table 15–307 describes how to use TxEn bit in order to achieve hardware flow control, it is strongly suggested to let UART1 hardware implemented auto flow control features take care of this, and limit the scope of TxEn to software flow control.

U1TER enables implementation of software and hardware flow control. When TXEn=1, UART1 transmitter will keep sending data as long as they are available. As soon as TXEn becomes 0, UART1 transmission will stop.

Table 15–307 describes how to use TXEn bit in order to achieve software flow control.

Table 307: UART1 Transmit Enable Register (U1TER - address 0x4001 0030) bit description

Bit	Symbol	Description	Reset Value
6:0	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7	TXEN	When this bit is 1, as it is after a Reset, data written to the THR is output on the TXD pin as soon as any preceding data has been sent. If this bit cleared to 0 while a character is being sent, the transmission of that character is completed, but no further characters are sent until this bit is set again. In other words, a 0 in this bit blocks the transfer of characters from the THR or TX FIFO into the transmit shift register. Software can clear this bit when it detects that the a hardware-handshaking TX-permit signal (CTS) has gone false, or with software handshaking, when it receives an XOFF character (DC3). Software can set this bit again when it detects that the TX-permit signal has gone true, or when it receives an XON (DC1) character.	1
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

4.18 UART1 RS485 Control register (U1RS485CTRL - 0x4001 004C)

The U1RS485CTRL register controls the configuration of the UART in RS-485/EIA-485 mode.

Table 308: UART1 RS485 Control register (U1RS485CTRL - address 0x4001 004C) bit description

Bit	Symbol	Value	Description	Reset value
0	NMMEN	0	RS-485/EIA-485 Normal Multidrop Mode (NMM) is disabled.	0
		1	RS-485/EIA-485 Normal Multidrop Mode (NMM) is enabled. In this mode, an address is detected when a received byte causes the UART to set the parity error and generate an interrupt.	
1	RXDIS	0	The receiver is enabled.	0
		1	The receiver is disabled.	
2	AADEN	0	Auto Address Detect (AAD) is disabled.	0
		1	Auto Address Detect (AAD) is enabled.	
3	SEL	0	If direction control is enabled (bit DCTRL = 1), pin $\overline{\text{RTS}}$ is used for direction control.	0
		1	If direction control is enabled (bit DCTRL = 1), pin DTR is used for direction control.	
4	DCTRL	0	Disable Auto Direction Control.	0
		1	Enable Auto Direction Control.	

Table 308: UART1 RS485 Control register (U1RS485CTRL - address 0x4001 004C) bit description

Bit	Symbol	Value	Description	Reset value
5	OINV		This bit reverses the polarity of the direction control signal on the RTS (or DTR) pin.	0
		0	The direction control pin will be driven to logic '0' when the transmitter has data to be sent. It will be driven to logic '1' after the last bit of data has been transmitted.	
		1	The direction control pin will be driven to logic '1' when the transmitter has data to be sent. It will be driven to logic '0' after the last bit of data has been transmitted.	
31:6	-	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

4.19 UART1 RS-485 Address Match register (U1RS485ADRMATCH - 0x4001 0050)

The U1RS485ADRMATCH register contains the address match value for RS-485/EIA-485 mode.

Table 309. UART1 RS-485 Address Match register (U1RS485ADRMATCH - address 0x4001 0050) bit description

Bit	Symbol	Description	Reset value
7:0	ADRMATCH	Contains the address match value.	0x00
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

4.20 UART1 RS-485 Delay value register (U1RS485DLY - 0x4001 0054)

The user may program the 8-bit RS485DLY register with a delay between the last stop bit leaving the TXFIFO and the de-assertion of $\overline{RTS}$ (or $\overline{DTR}$). This delay time is in periods of the baud clock. Any delay time from 0 to 255 bit times may be programmed.

Table 310. UART1 RS-485 Delay value register (U1RS485DLY - address 0x4001 0054) bit description

Bit	Symbol	Description	Reset value
7:0	DLY	Contains the direction control (RTS or DTR) delay value. This register works in conjunction with an 8-bit counter.	0x00
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

4.21 RS-485/EIA-485 modes of operation

The RS-485/EIA-485 feature allows the UART to be configured as an addressable slave. The addressable slave is one of multiple slaves controlled by a single master.

The UART master transmitter will identify an address character by setting the parity (9th) bit to '1'. For data characters, the parity bit is set to '0'.

Each UART slave receiver can be assigned a unique address. The slave can be programmed to either manually or automatically reject data following an address which is not theirs.

RS-485/EIA-485 Normal Multidrop Mode (NMM)

Setting the RS485CTRL bit 0 enables this mode. In this mode, an address is detected when a received byte causes the UART to set the parity error and generate an interrupt.

If the receiver is DISABLED (RS485CTRL bit 1 = ‘1’) any received data bytes will be ignored and will not be stored in the RXFIFO. When an address byte is detected (parity bit = ‘1’) it will be placed into the RXFIFO and an Rx Data Ready Interrupt will be generated. The processor can then read the address byte and decide whether or not to enable the receiver to accept the following data.

While the receiver is ENABLED (RS485CTRL bit 1 =’0’) all received bytes will be accepted and stored in the RXFIFO regardless of whether they are data or address. When an address character is received a parity error interrupt will be generated and the processor can decide whether or not to disable the receiver.

RS-485/EIA-485 Auto Address Detection (AAD) mode

When both RS485CTRL register bits 0 (9-bit mode enable) and 2 (AAD mode enable) are set, the UART is in auto address detect mode.

In this mode, the receiver will compare any address byte received (parity = ‘1’) to the 8-bit value programmed into the RS485ADRMATCH register.

If the receiver is DISABLED (RS485CTRL bit 1 = ‘1’) any received byte will be discarded if it is either a data byte OR an address byte which fails to match the RS485ADRMATCH value.

When a matching address character is detected it will be pushed onto the RXFIFO along with the parity bit, and the receiver will be automatically enabled (RS485CTRL bit 1 will be cleared by hardware). The receiver will also generate n Rx Data Ready Interrupt.

While the receiver is ENABLED (RS485CTRL bit 1 = ‘0’) all bytes received will be accepted and stored in the RXFIFO until an address byte which does not match the RS485ADRMATCH value is received. When this occurs, the receiver will be automatically disabled in hardware (RS485CTRL bit 1 will be set), The received non-matching address character will not be stored in the RXFIFO.

RS-485/EIA-485 Auto Direction Control

RS485/EIA-485 Mode includes the option of allowing the transmitter to automatically control the state of either the RTS pin or the DTR pin as a direction control output signal.

Setting RS485CTRL bit 4 = ‘1’ enables this feature.

Direction control, if enabled, will use the RTS pin when RS485CTRL bit 3 = ‘0’. It will use the DTR pin when RS485CTRL bit 3 = ‘1’.

When Auto Direction Control is enabled, the selected pin will be asserted (driven low) when the CPU writes data into the TXFIFO. The pin will be de-asserted (driven high) once the last bit of data has been transmitted. See bits 4 and 5 in the RS485CTRL register.

The RS485CTRL bit 4 takes precedence over all other mechanisms controlling RTS (or DTR) with the exception of loopback mode.

RS485/EIA-485 driver delay time

The driver delay time is the delay between the last stop bit leaving the TXFIFO and the de-assertion of RTS (or DTR). This delay time can be programmed in the 8-bit RS485DLY register. The delay time is in periods of the baud clock. Any delay time from 0 to 255 bit times may be programmed.

RS485/EIA-485 output inversion

The polarity of the direction control signal on the $\overline{RTS}$ (or $\overline{DTR}$) pins can be reversed by programming bit 5 in the U1RS485CTRL register. When this bit is set, the direction control pin will be driven to logic 1 when the transmitter has data waiting to be sent. The direction control pin will be driven to logic 0 after the last bit of data has been transmitted.

4.22 UART1 FIFO Level register (U1FIFOLVL - 0x4001 0058)

U1FIFOLVL register is a read-only register that allows software to read the current FIFO level status. Both the transmit and receive FIFO levels are present in this register.

Table 311. UART1 FIFO Level register (U1FIFOLVL - address 0x4001 0058) bit description

Bit	Symbol	Description	Reset value
3:0	RXFIFILVL	Reflects the current level of the UART1 receiver FIFO. 0 = empty, 0xF = FIFO full.	0x00
7:4	-	Reserved. The value read from a reserved bit is not defined.	NA
11:8	TXFIFOLVL	Reflects the current level of the UART1 transmitter FIFO. 0 = empty, 0xF = FIFO full.	0x00
31:12	-	Reserved, the value read from a reserved bit is not defined.	NA

5. Architecture

The architecture of the UART1 is shown below in the block diagram.

The APB interface provides a communications link between the CPU or host and the UART1.

The UART1 receiver block, U1RX, monitors the serial input line, RXD1, for valid input. The UART1 RX Shift Register (U1RSR) accepts valid characters via RXD1. After a valid character is assembled in the U1RSR, it is passed to the UART1 RX Buffer Register FIFO to await access by the CPU or host via the generic host interface.

The UART1 transmitter block, U1TX, accepts data written by the CPU or host and buffers the data in the UART1 TX Holding Register FIFO (U1THR). The UART1 TX Shift Register (U1TSR) reads the data stored in the U1THR and assembles the data to transmit via the serial output pin, TXD1.

The UART1 Baud Rate Generator block, U1BRG, generates the timing enables used by the UART1 TX block. The U1BRG clock input source is the APB clock (PCLK). The main clock is divided down per the divisor specified in the U1DLL and U1DLM registers. This divided down clock is a 16x oversample clock, NBAUDOUT.

The modem interface contains registers U1MCR and U1MSR. This interface is responsible for handshaking between a modem peripheral and the UART1.

The interrupt interface contains registers U1IER and U1IIR. The interrupt interface receives several one clock wide enables from the U1TX and U1RX blocks.

Status information from the U1TX and U1RX is stored in the U1LSR. Control information for the U1TX and U1RX is stored in the U1LCR.

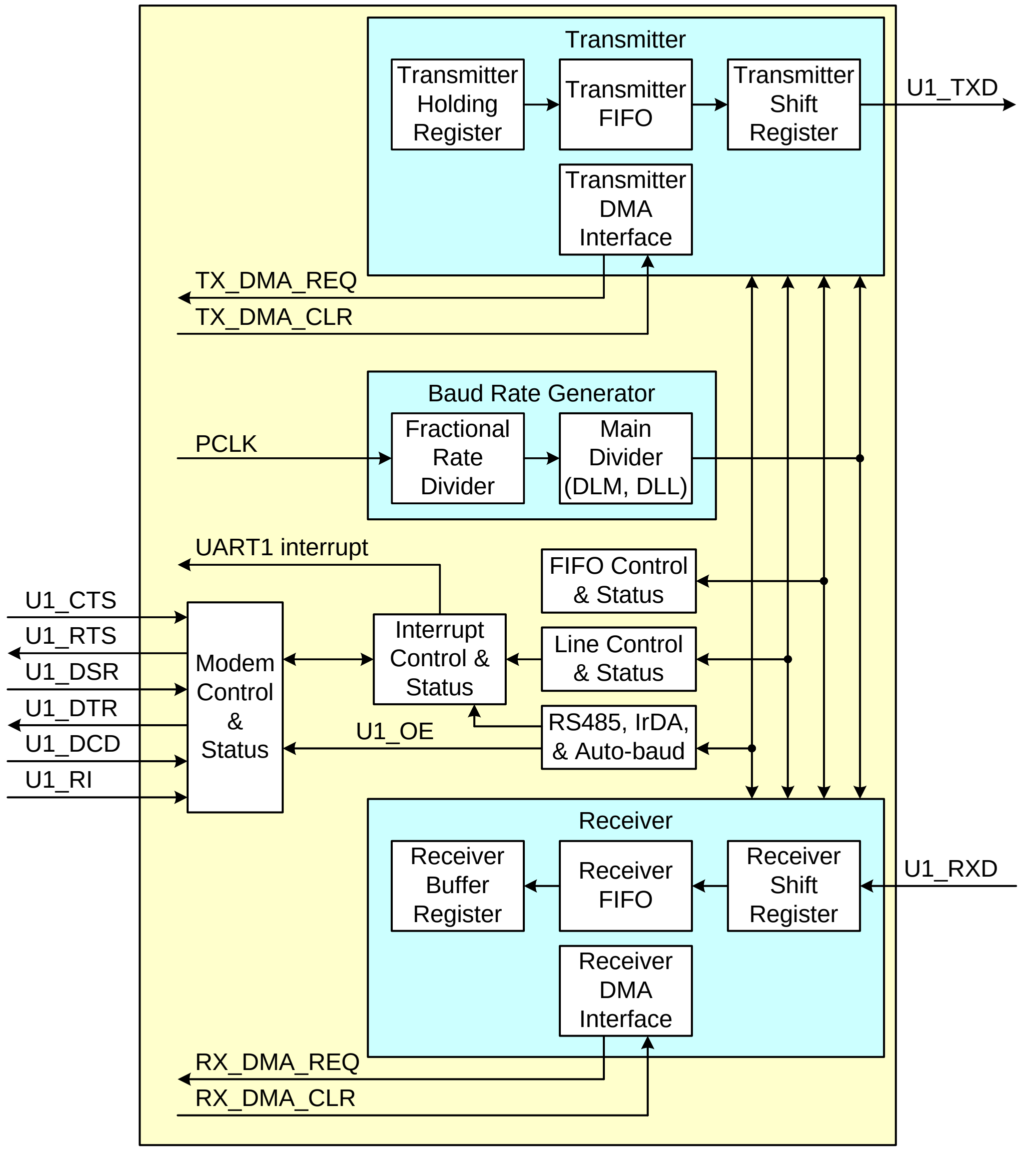


Fig 51. UART1 block diagram